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(54) REFRIGERATOR AND METHOD FOR CONTROLLING THE SAME

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(57) ABSTRACT

A refrigerator including a main body including a storage compartment; a first cooler configured to cool the storage compartment, and including a thermoelectric element, a heat dissipation fan, and a cooling fan; a second cooler configured to cool the storage compartment, and including a compressor, and an evaporator fan; and at least one processor configured to, based on a failure in the first cooler, increase an operating speed of the compressor and the evaporator fan of the second cooler.

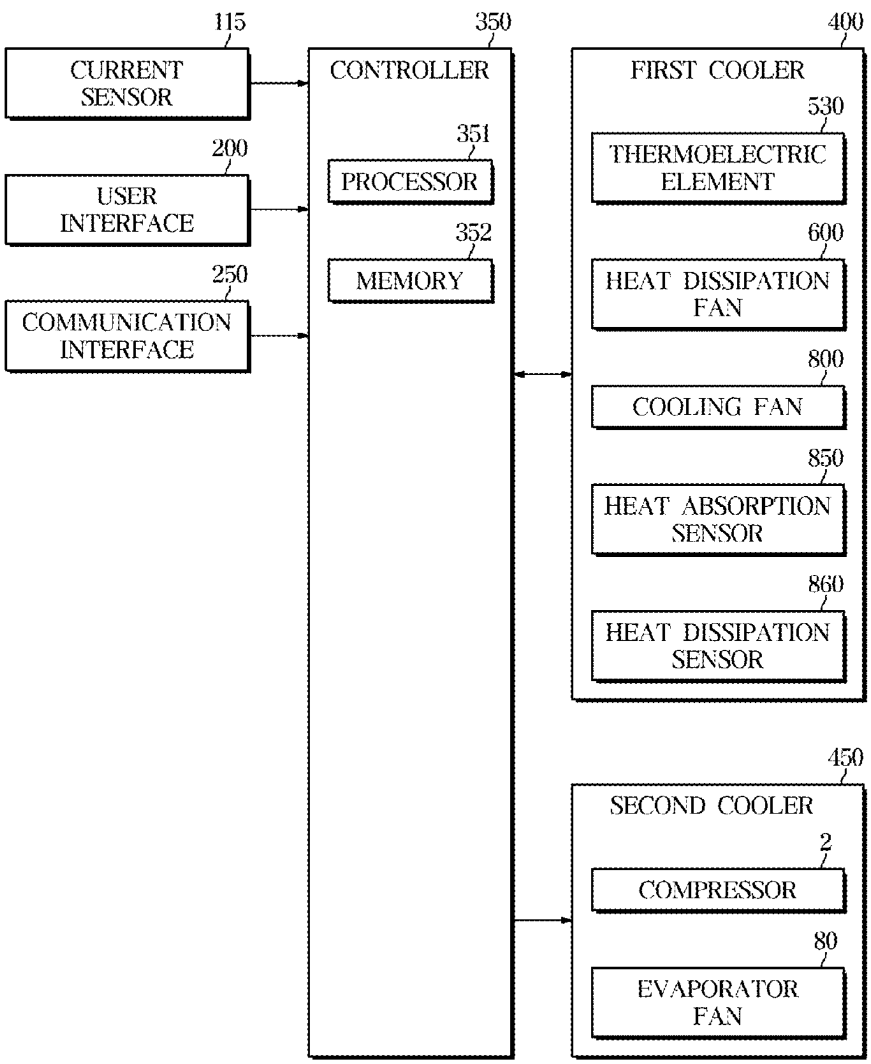


FIG. 1

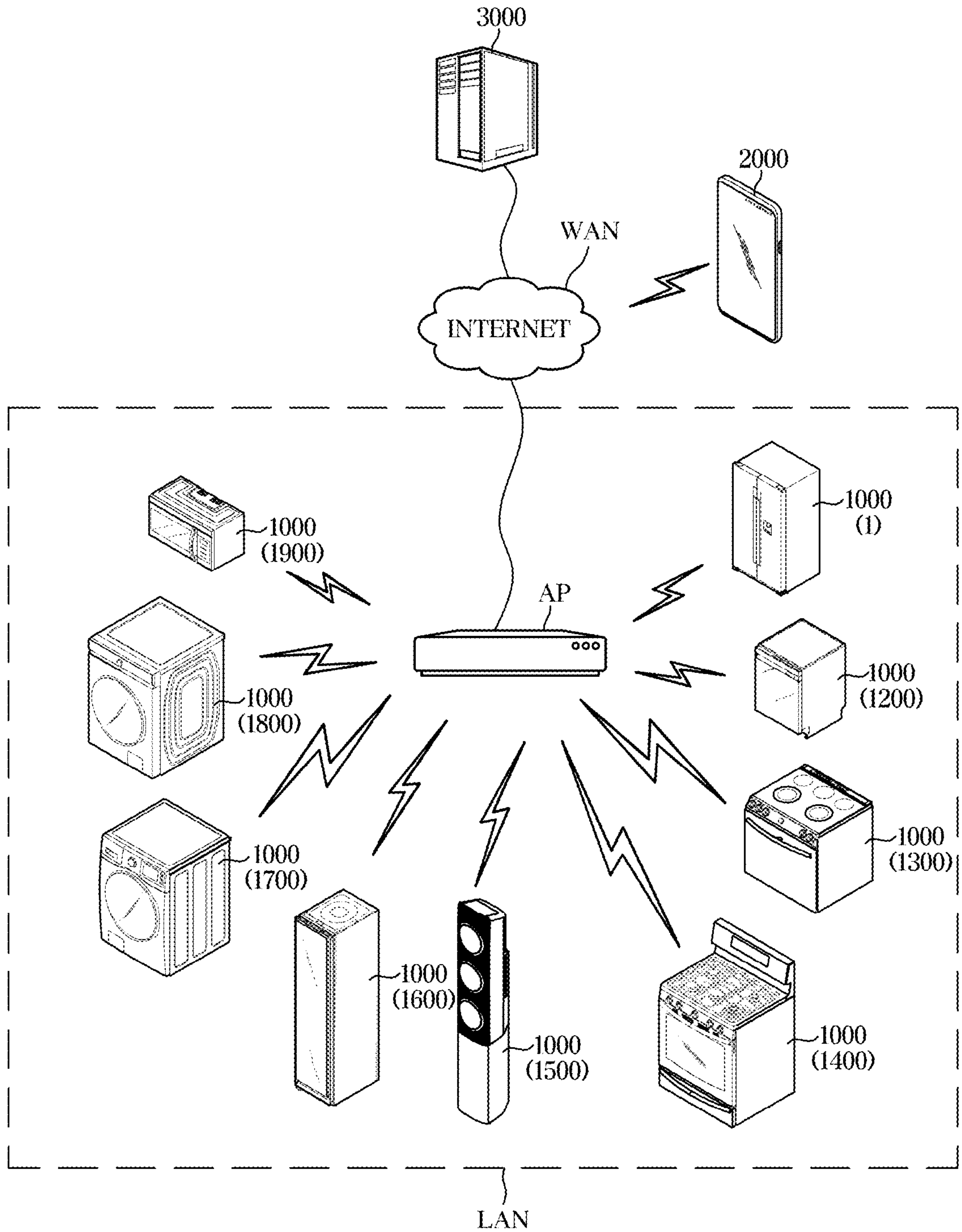

3000
2000
WAN
INTERNET
AP
1000
(1900)
1000
(1)
1000
(1800)
1000
(1200)
1000
(1700)
1000
(1300)
1000
(1600)
1000
(1500)
1000
(1400)
LAN

FIG. 2

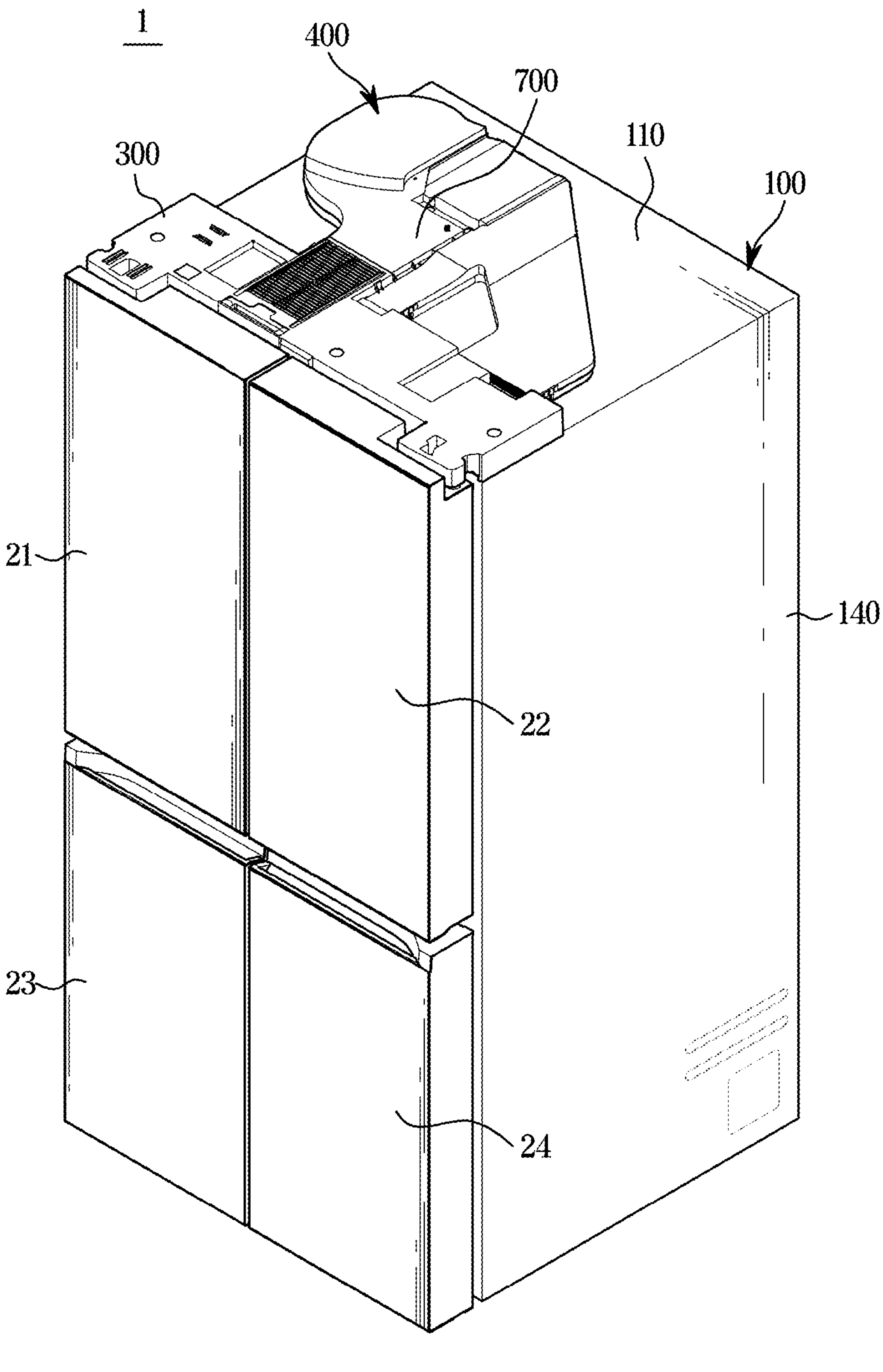
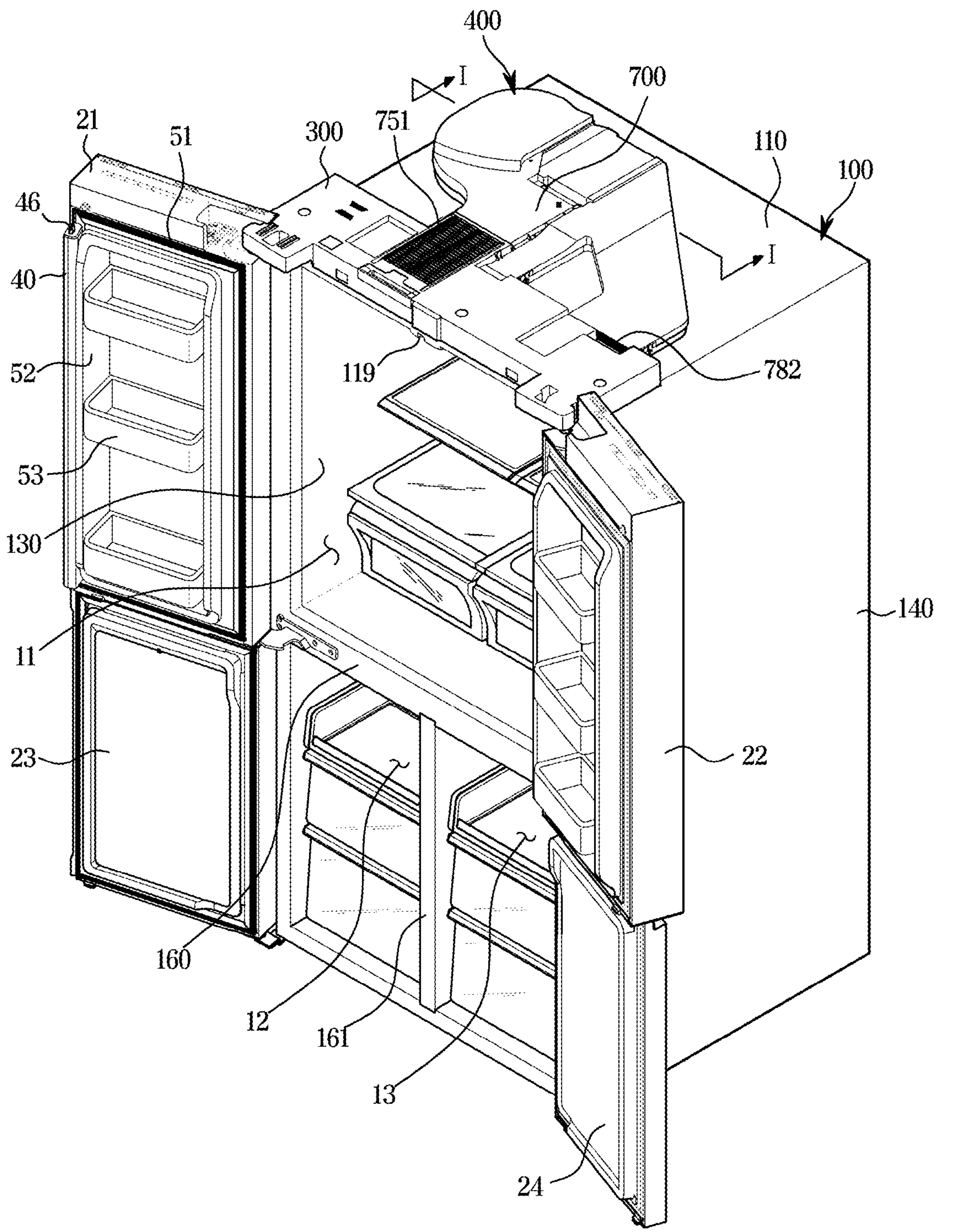


FIG. 3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I
21
51
300
751
110
100
46
I
40
52
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53
130
140
11
23
22
160
12
161
13
24

FIG. 4

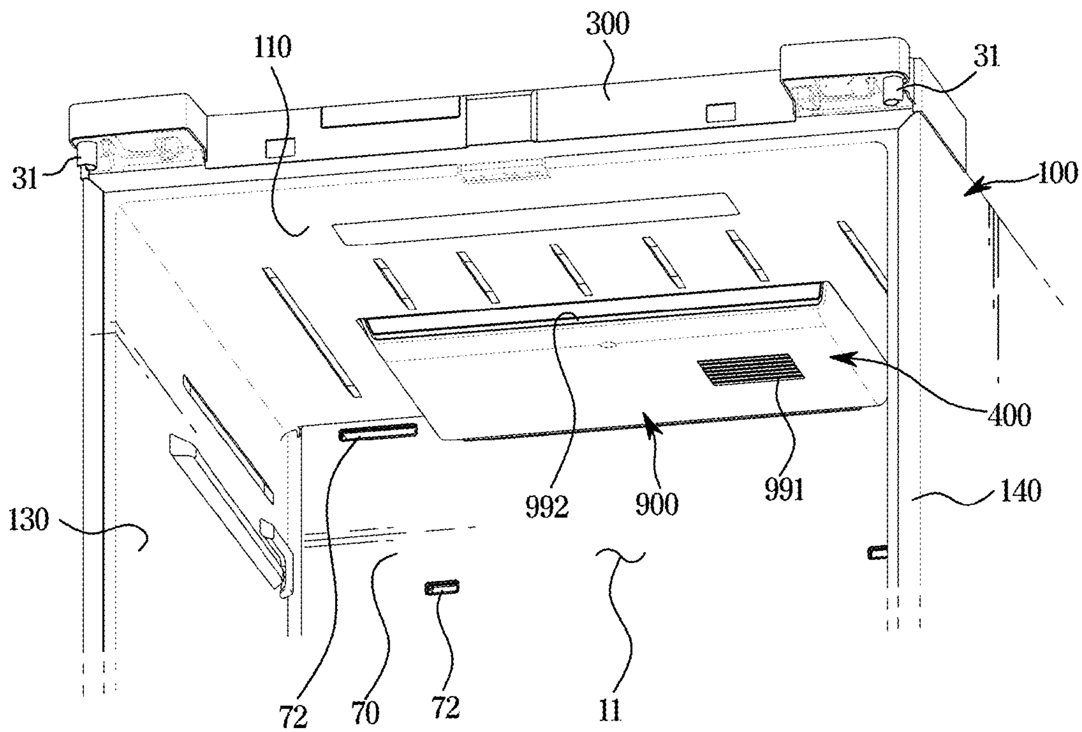


FIG. 5

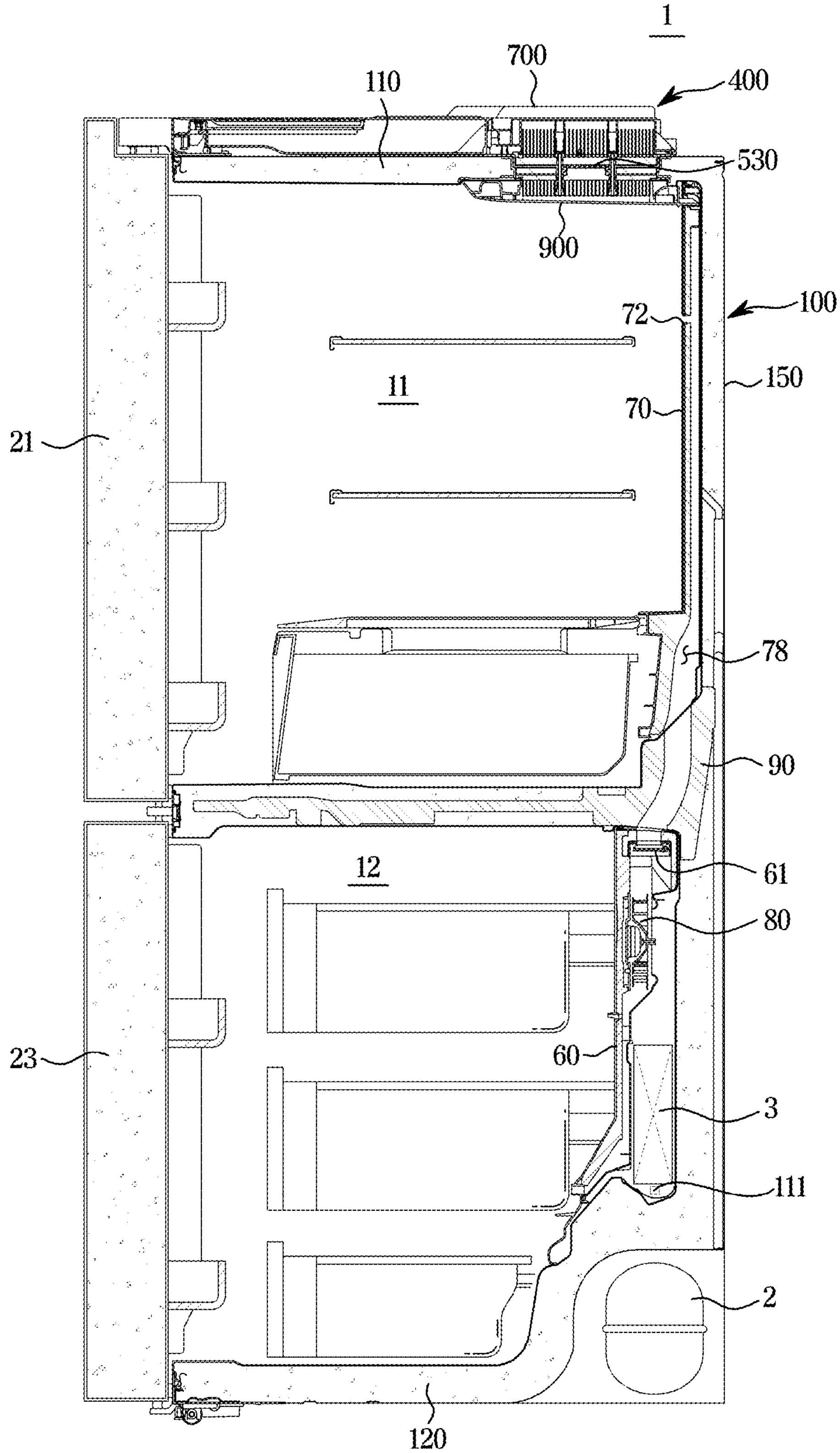
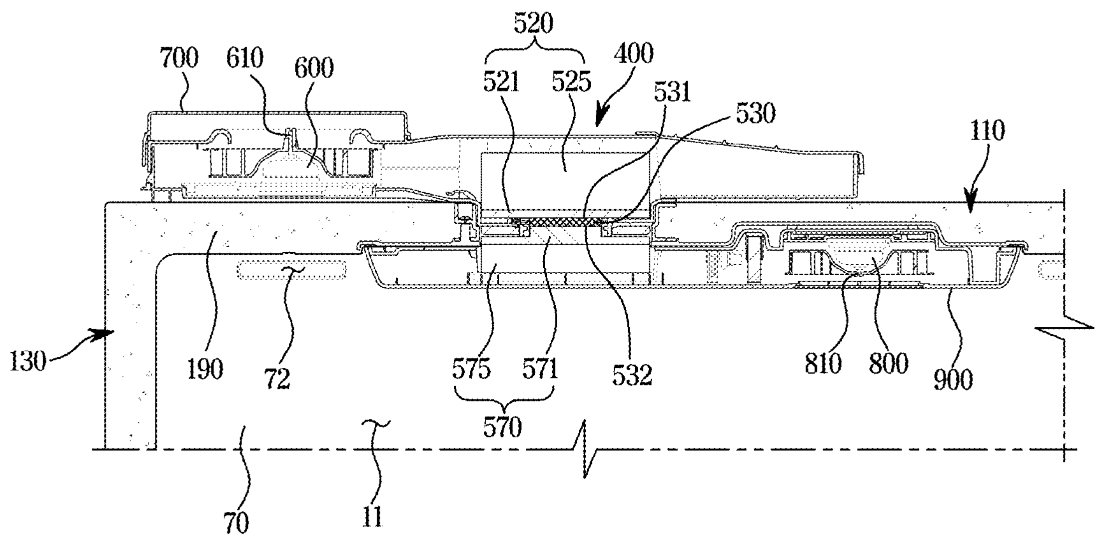

1
700
110
400
530
900
100
72
150
11
70
21
78
90
12
61
80
60
23
3
111
2
120

FIG. 6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700
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521
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530
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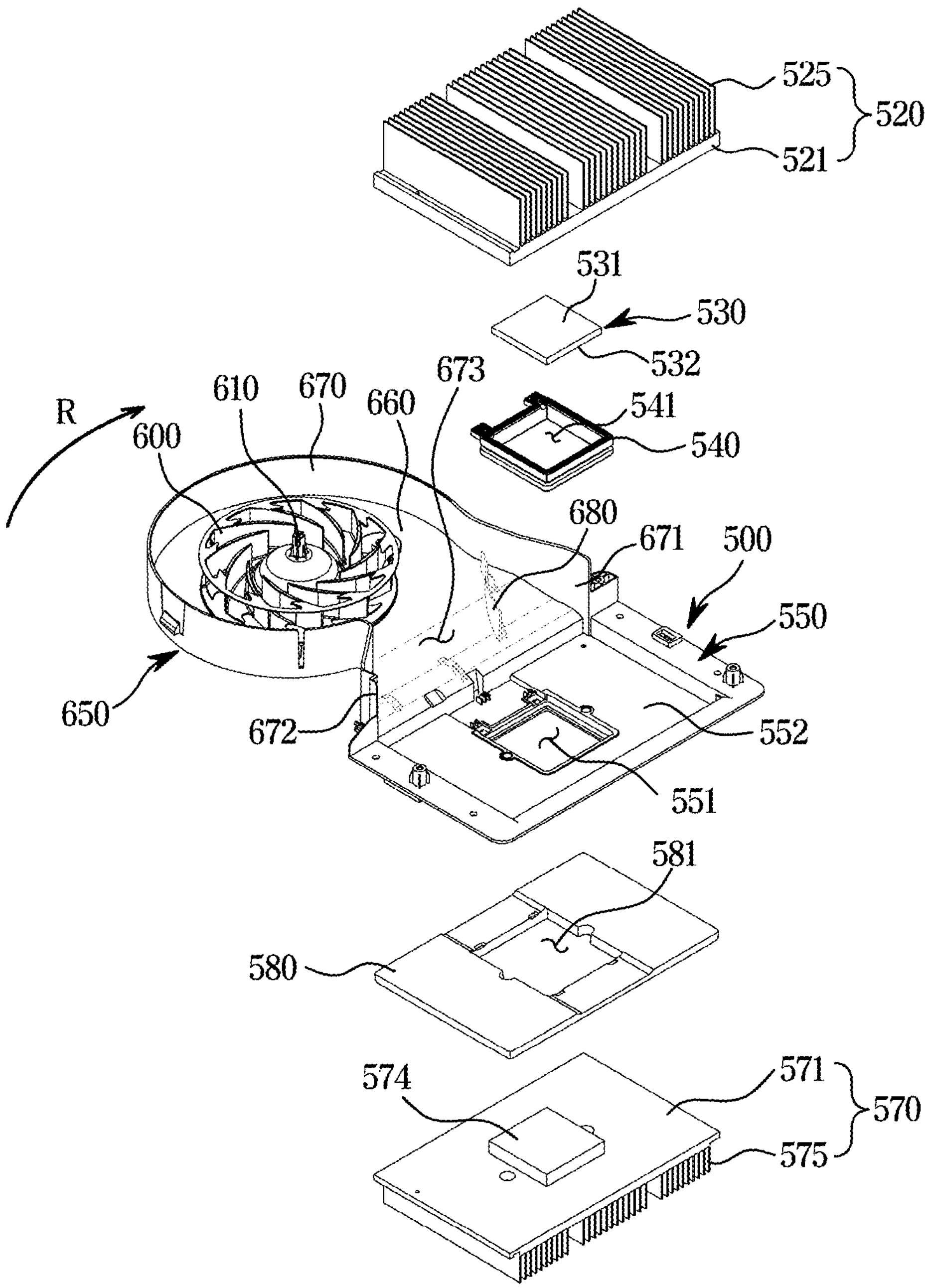
72
575
571
532
810
800
900
570
70
11

FIG. 7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610
670
660
541
R
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672
552
551
581
580
574
571
570
575

FIG. 8

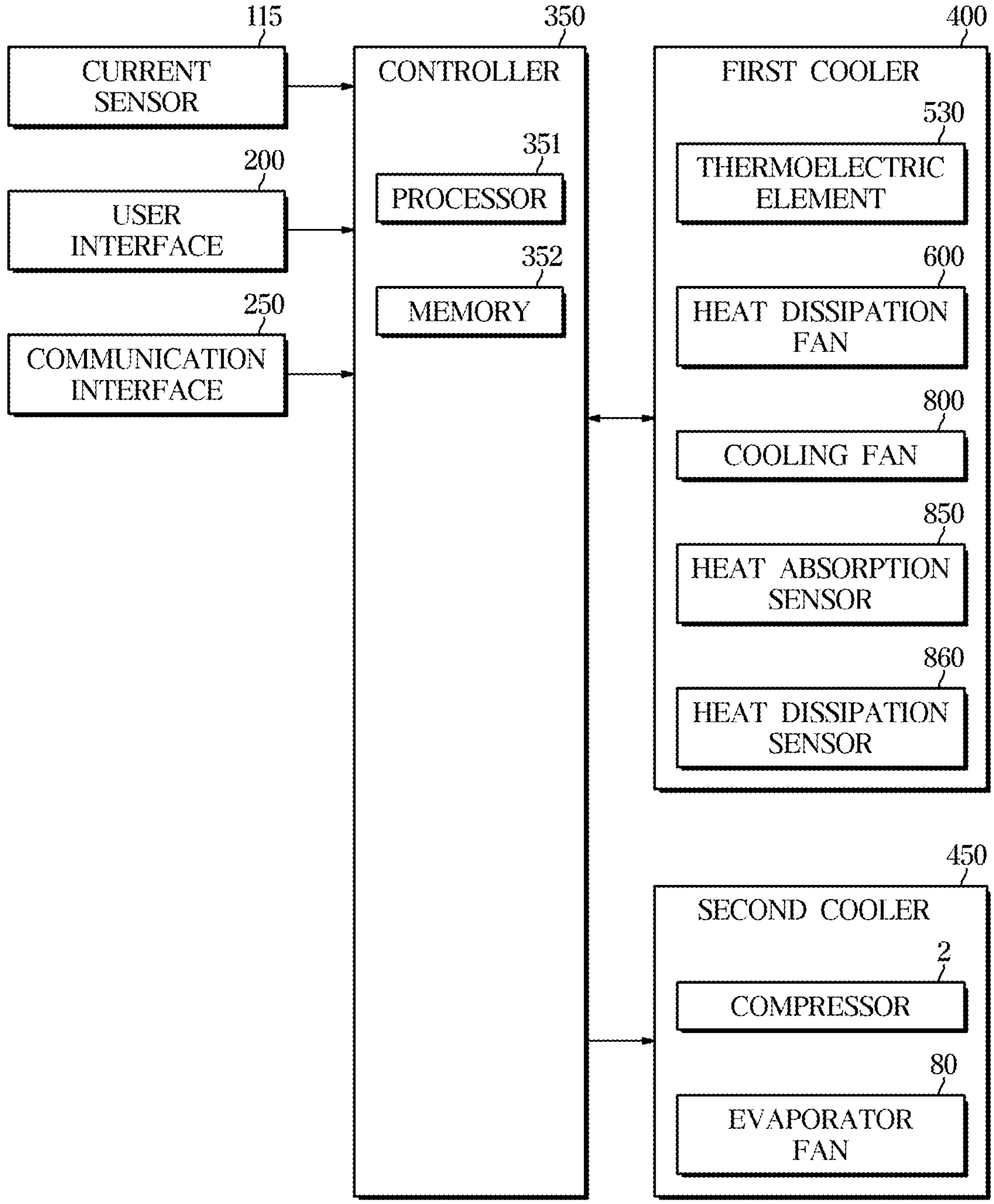

115
CURRENT SENSOR
200
USER INTERFACE
250
COMMUNICATION INTERFACE
350
CONTROLLER
351
PROCESSOR
352
MEMORY
400
FIRST COOLER
530
THERMOELECTRIC ELEMENT
600
HEAT DISSIPATION FAN
800
COOLING FAN
850
HEAT ABSORPTION SENSOR
860
HEAT DISSIPATION SENSOR
450
SECOND COOLER
2
COMPRESSOR
80
EVAPORATOR FAN

FIG. 9

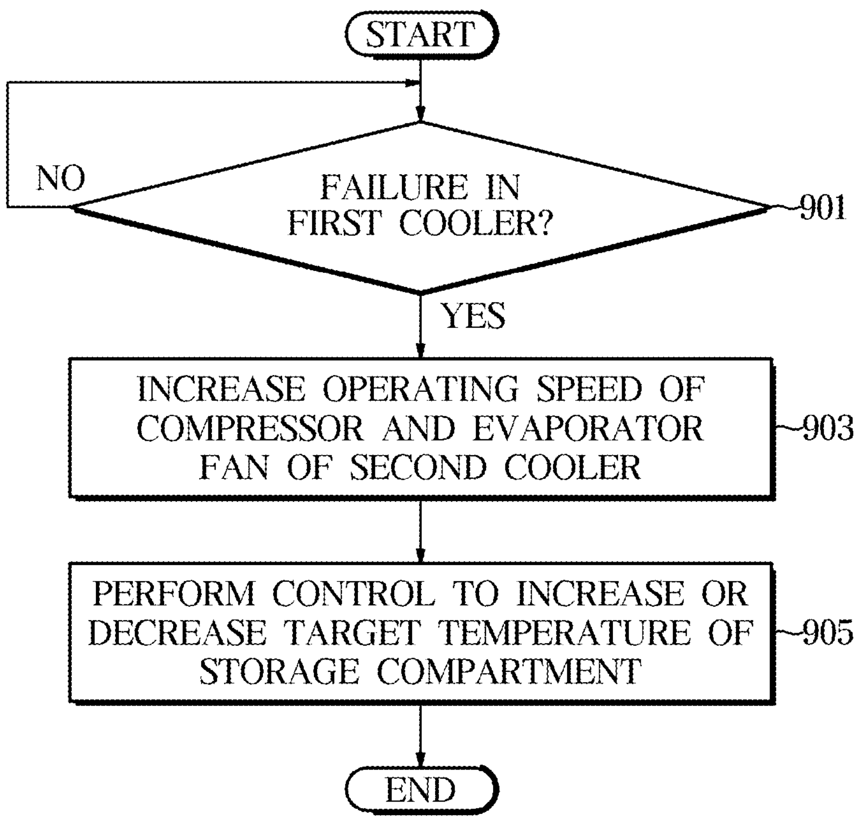

START
NO
FAILURE IN FIRST COOLER?
901
YES
INCREASE OPERATING SPEED OF COMPRESSOR AND EVAPORATOR FAN OF SECOND COOLER
903
PERFORM CONTROL TO INCREASE OR DECREASE TARGET TEMPERATURE OF STORAGE COMPARTMENT
905
END

FIG. 10

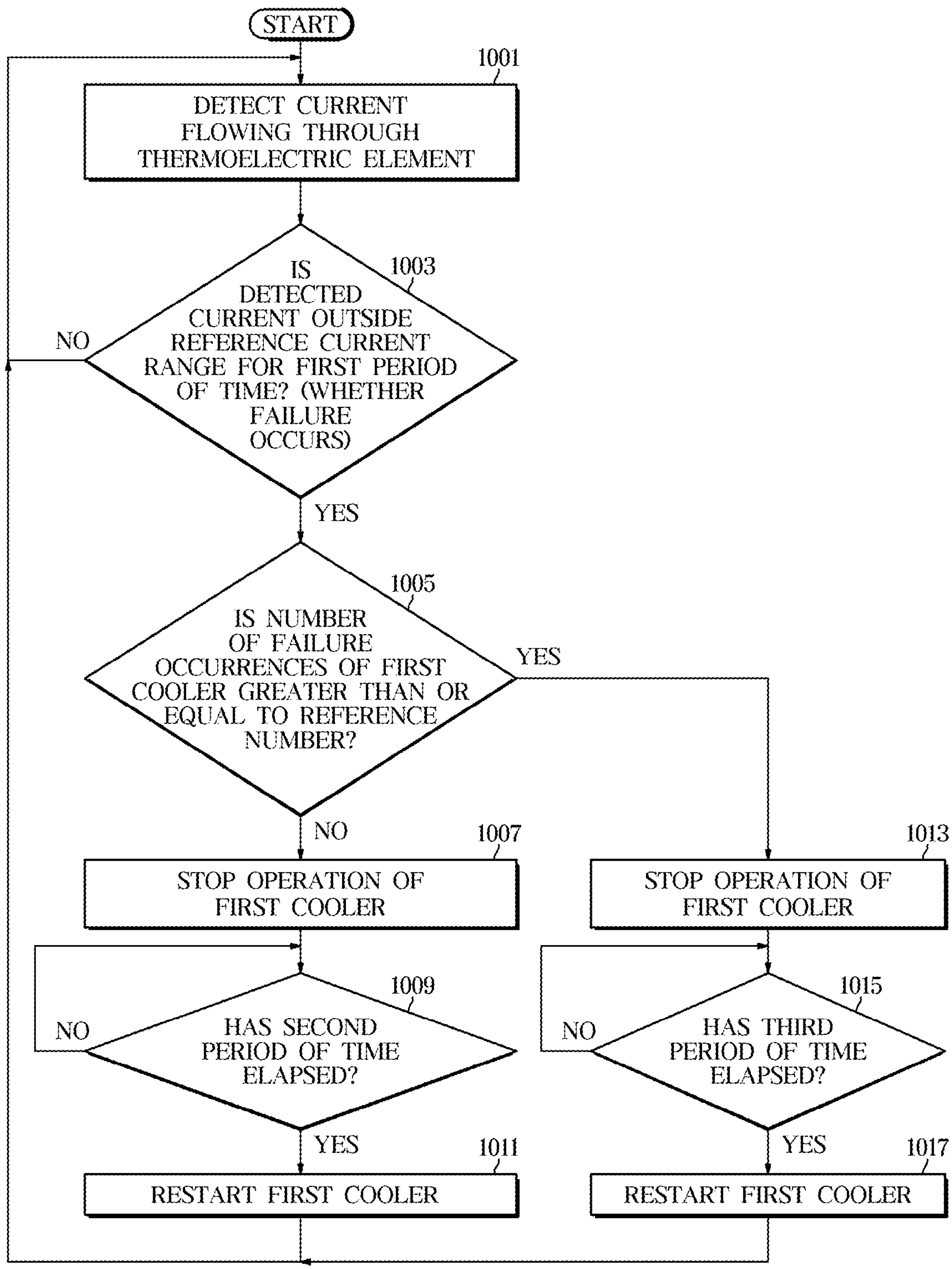

START
1001
DETECT CURRENT FLOWING THROUGH THERMOELECTRIC ELEMENT
1003
IS DETECTED CURRENT OUTSIDE REFERENCE CURRENT RANGE FOR FIRST PERIOD OF TIME? (WHETHER FAILURE OCCURS)
NO
YES
1005
IS NUMBER OF FAILURE OCCURRENCES OF FIRST COOLER GREATER THAN OR EQUAL TO REFERENCE NUMBER?
YES
NO
1007
STOP OPERATION OF FIRST COOLER
1013
STOP OPERATION OF FIRST COOLER
1009
HAS SECOND PERIOD OF TIME ELAPSED?
NO
YES
1015
HAS THIRD PERIOD OF TIME ELAPSED?
NO
YES
1011
RESTART FIRST COOLER
1017
RESTART FIRST COOLER

FIG. 11

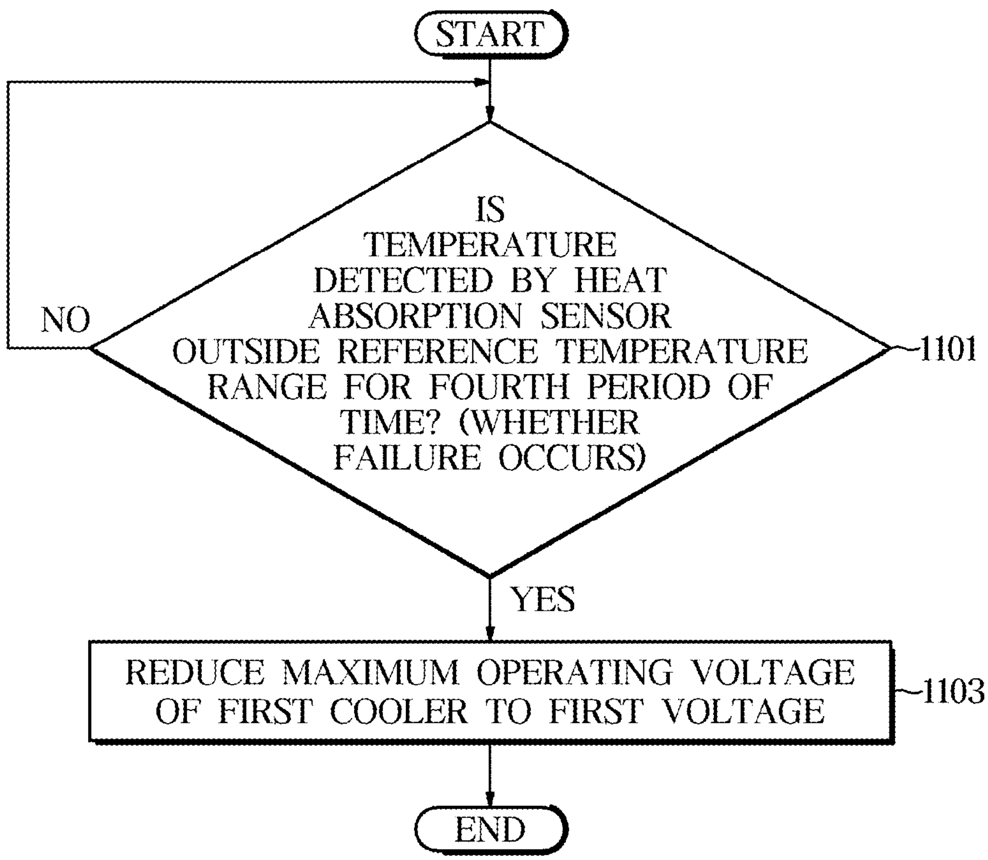

START
IS TEMPERATURE DETECTED BY HEAT ABSORPTION SENSOR OUTSIDE REFERENCE TEMPERATURE RANGE FOR FOURTH PERIOD OF TIME? (WHETHER FAILURE OCCURS)
NO
1101
YES
REDUCE MAXIMUM OPERATING VOLTAGE OF FIRST COOLER TO FIRST VOLTAGE
1103
END

FIG. 12

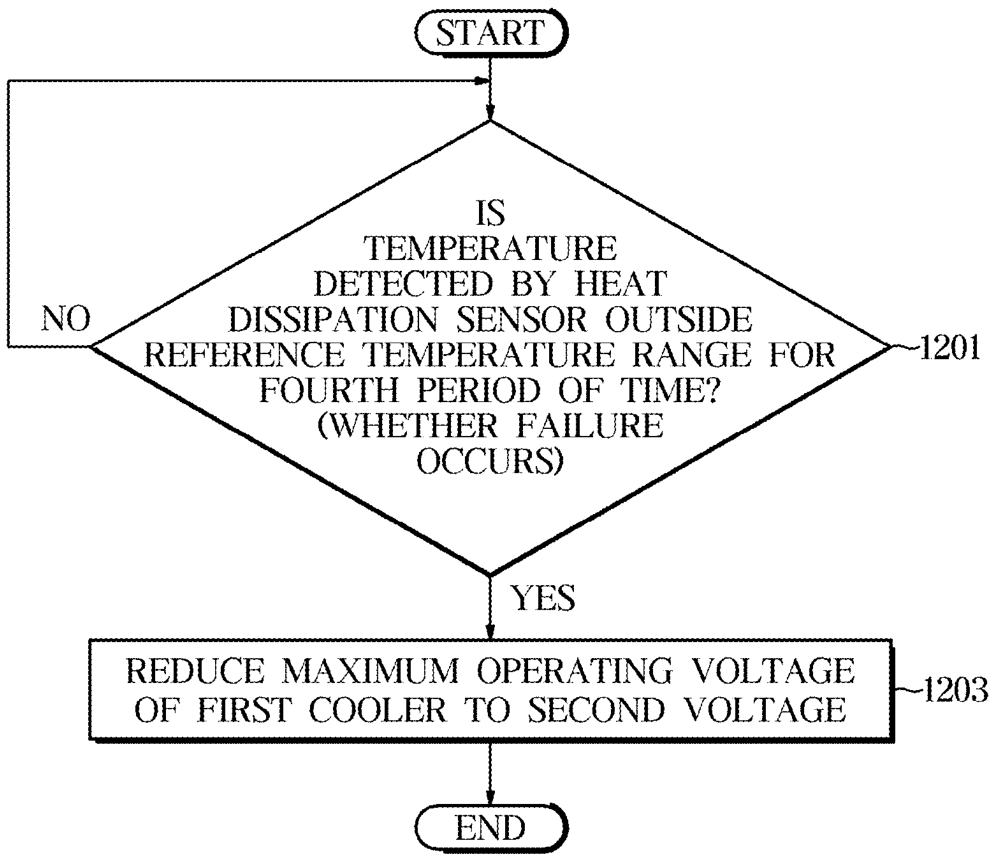
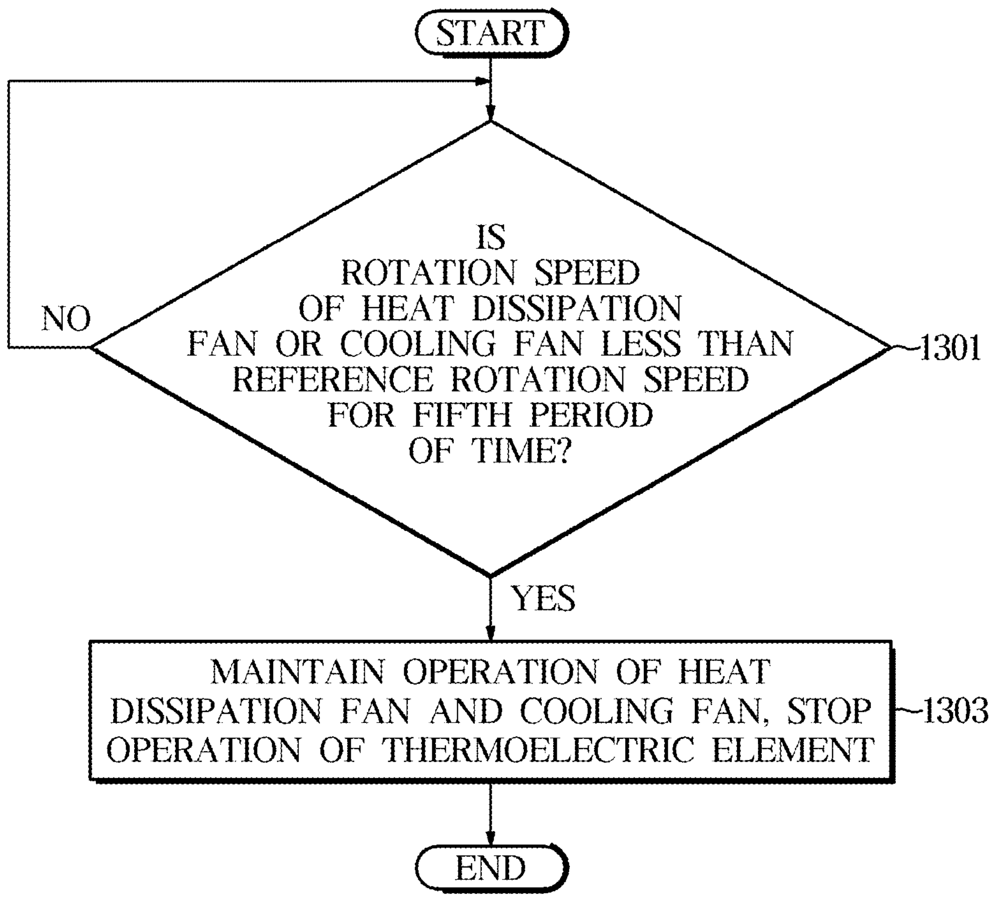

START
IS TEMPERATURE DETECTED BY HEAT DISSIPATION SENSOR OUTSIDE REFERENCE TEMPERATURE RANGE FOR FOURTH PERIOD OF TIME? (WHETHER FAILURE OCCURS)
NO
1201
YES
REDUCE MAXIMUM OPERATING VOLTAGE OF FIRST COOLER TO SECOND VOLTAGE
1203
END

FIG. 13


START
IS ROTATION SPEED OF HEAT DISSIPATION FAN OR COOLING FAN LESS THAN REFERENCE ROTATION SPEED FOR FIFTH PERIOD OF TIME?
NO
1301
YES
MAINTAIN OPERATION OF HEAT DISSIPATION FAN AND COOLING FAN, STOP OPERATION OF THERMOELECTRIC ELEMENT
1303
END

REFRIGERATOR AND METHOD FOR CONTROLLING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation application of International Application No. PCT/KR2025/001078, filed on Jan. 20, 2025, which is based on and claims the benefit of Korean Patent Application Number 10-2024-0032701, filed on Mar. 7, 2024, and Korean Patent Application Number 10-2024-0103427, filed on Aug. 2, 2024, the disclosures of which are incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] The disclosure relates to a refrigerator including a thermoelectric element and a compressor to cool a storage compartment, and a method for controlling the refrigerator.

BACKGROUND ART

[0003] A refrigerator is an appliance that is equipped with a main body having a storage compartment and a cold air supply device for supplying the storage compartment with cold air to store food in a fresh state.

[0004] A thermoelectric cooler that generates heating and cooling by the Peltier effect may be used as a cold air supply device of a refrigerator. The thermoelectric cooler may include a thermoelectric element. The thermoelectric element has a heating portion formed on one side and a cooling portion formed on the opposite side, and when current is applied to the thermoelectric element, heating may occur in the heating portion and heat absorption may occur in the cooling portion.

[0005] The thermoelectric cooler may include a heat sink, a cooling sink, a heat dissipation fan, a cooling fan, a heat dissipation duct, and a cooling duct to increase the cooling efficiency of a storage compartment through the thermoelectric cooler.

DISCLOSURE

Technical Problem

[0006] The disclosure provides a refrigerator that may maintain a proper temperature of the refrigerator by detecting various types of failure of a thermoelectric cooler and performing control based on the type of failure, and a method for controlling the same.

[0007] The disclosure provides a refrigerator that may maintain a temperature of the refrigerator by increasing an operating intensity of a refrigeration cycle device in response to a failure of a thermoelectric cooler.

[0008] Technical objects that can be achieved by the disclosure are not limited to the above-mentioned objects, and other technical objects not mentioned will be clearly understood by one of ordinary skill in the art to which the disclosure belongs from the following description.

Technical Solution

[0009] Aspects of embodiments of the disclosure will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0010] According to an embodiment of the disclosure, a refrigerator includes a main body including a storage compartment; a first cooler configured to cool the storage compartment, and including a thermoelectric element, a heat dissipation fan, and a cooling fan; a second cooler configured to cool the storage compartment, and including a compressor, and an evaporator fan; and at least one processor configured to, based on a failure in the first cooler, increase an operating speed of the compressor and the evaporator fan of the second cooler.

[0011] According to an embodiment of the disclosure, the at least one processor may be configured to, based on the failure in the first cooler, increase or decrease a target temperature of the storage compartment.

[0012] According to an embodiment of the disclosure, the refrigerator may further include a current sensor configured to detect a current flowing through the thermoelectric element. The failure in the first cooler may include the current detected by the current sensor being outside a reference current range for a predetermined first period of time.

[0013] According to an embodiment of the disclosure, the at least one processor may be configured to, based on the current detected by the current sensor being outside the reference current range for the predetermined first period of time, stop an operation of the first cooler, and, based on an elapse of a predetermined second period of time, restart the operation of the first cooler.

[0014] According to an embodiment of the disclosure, the at least one processor may be configured to, based on the failure in the first cooler occurring more than a reference number of times, stop the operation of the first cooler for a predetermined third period of time that is longer than the predetermined second period of time.

[0015] According to an embodiment of the disclosure, the refrigerator may further include a voltage sensor configured to detect a voltage applied to the thermoelectric element. The failure in the first cooler may include the voltage detected by the voltage sensor being outside a reference voltage range for a predetermined first period of time.

[0016] According to an embodiment of the disclosure, the at least one processor may be configured to, based on the voltage detected by the voltage sensor being outside the reference voltage range for the predetermined first period of time, stop an operation of the first cooler, and, based on an elapse of a predetermined second period of time, restart the operation of the first cooler.

[0017] According to an embodiment of the disclosure, the first cooler may include a heat absorption sensor configured to detect a temperature of air absorbed by the thermoelectric element, and a heat dissipation sensor configured to detect a temperature of air discharged by the thermoelectric element. The failure in the first cooler may include the temperature detected by the heat absorption sensor or the heat dissipation sensor being outside a reference temperature range for a predetermined period of time.

[0018] According to an embodiment of the disclosure, the at least one processor may be configured to, based on the temperature detected by the heat absorption sensor or the heat dissipation sensor being outside the reference temperature range for the predetermined period of time, reduce a maximum operating voltage of the first cooler.

[0019] According to an embodiment of the disclosure, the at least one processor may be configured to, based on the temperature detected by the heat absorption sensor being outside the reference temperature range for the predetermined period of time, reduce the maximum operating voltage of the first cooler to a first voltage, and, based on the temperature detected by the heat dissipation sensor being outside the reference temperature range for the predetermined period of time, reduce the maximum operating voltage of the first cooler to a second voltage.

[0020] According to an embodiment of the disclosure, the failure in the first cooler may include a rotation speed of the heat dissipation fan or the cooling fan being less than a reference rotation speed for a predetermined period of time.

[0021] According to an embodiment of the disclosure, the at least one processor may be configured to, based on the rotation speed of the heat dissipation fan or the cooling fan being less than the reference rotation speed for the predetermined period of time, maintain an operation of the heat dissipation fan or the cooling fan, and stop an operation of the thermoelectric element.

[0022] According to an embodiment of the disclosure, provided is a method of controlling a refrigerator including a main body including a storage compartment, a first cooler configured to cool the storage compartment and including a thermoelectric element, a heat dissipation fan, and a cooling fan, and a second cooler configured to cool the storage compartment and including a compressor and an evaporator fan, the method including detecting whether a failure occurs in the first cooler; and, based on the failure in the first cooler being detected, increasing an operating speed of the compressor and the evaporator fan of the second cooler.

[0023] According to an embodiment of the disclosure, the method may further include, based on the failure in the first cooler being detected, increasing or decreasing a target temperature of the storage compartment.

[0024] According to an embodiment of the disclosure, the refrigerator may include a current sensor configured to detect a current flowing through the thermoelectric element, and the failure in the first cooler being detected may include the current detected by the current sensor being outside a reference current range for a predetermined period of time.

DESCRIPTION OF DRAWINGS

[0025] These and/or other aspects of the disclosure will become apparent and more readily appreciated from the following description of embodiments, taken in conjunction with the accompanying drawings listed below.

[0026] FIG. **1** is a diagram illustrating communication among a home appliance, a server, a user terminal, and the like.

[0027] FIG. **2** illustrates a refrigerator according to an embodiment of the disclosure.

[0028] FIG. **3** is a view illustrating a state in which doors of a refrigerator are opened according to an embodiment of the disclosure.

[0029] FIG. **4** is a view illustrating an upper part of a storage compartment of a refrigerator, viewed from below, according to an embodiment of the disclosure.

[0030] FIG. **5** is a schematic side cross-sectional view of a refrigerator according to an embodiment of the disclosure.

[0031] FIG. **6** is a cross-sectional view along I-I' of FIG. **3**.

[0032] FIG. **7** is an exploded view of a thermoelectric cooler according to an embodiment of the disclosure.

[0033] FIG. **8** is a block diagram illustrating an example configuration of a refrigerator according to an embodiment of the disclosure.

[0034] FIG. **9** is a flowchart illustrating a method for controlling a refrigerator according to an embodiment of the disclosure.

[0035] FIG. **10** is a flowchart illustrating operations of a refrigerator depending on whether a failure has occurred in a thermoelectric cooler based on a current detection result according to an embodiment of the disclosure.

[0036] FIG. **11** and FIG. **12** are flowcharts illustrating operations of a refrigerator depending on whether a failure has occurred in a thermoelectric cooler based on a temperature detection result according to an embodiment of the disclosure.

[0037] FIG. **13** is a flowchart illustrating operations of a refrigerator depending on whether a failure has occurred in a thermoelectric cooler based on a fan rotation speed according to an embodiment of the disclosure.

MODES OF THE DISCLOSURE

[0038] Various embodiments of the disclosure and terms used herein are not intended to limit the technical features described herein to specific embodiments, and should be understood to include various modifications, equivalents, or substitutions of the corresponding embodiments.

[0039] In describing of the drawings, similar reference numerals may be used for similar or related elements.

[0040] The singular form of a noun corresponding to an item may include one or more of the items unless clearly indicated otherwise in a related context.

[0041] In the disclosure, phrases, such as "A or B", "at least one of A and B", "at least one of A or B", "A, B or C", "at least one of A, B and C", and "at least one of A, B, or C" may include any one or all possible combinations of the items listed together in the corresponding phrase among the phrases.

[0042] As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

[0043] Terms such as "1st", "2nd", "primary", or "secondary" may be used simply to distinguish an element from other elements, without limiting the element in other aspects (e.g., importance or order).

[0044] Further, as used in the disclosure, the terms "front", "rear", "top", "bottom", "side", "left", "right", "upper", "lower", and the like are defined with reference to the drawings, and are not intended to limit the shape and position of any element.

[0045] It will be understood that when the terms "includes", "comprises", "including", and/or "comprising" are used in the disclosure, they specify the presence of the specified features, figures, steps, operations, components, members, or combinations thereof, but do not preclude the presence or addition of one or more other features, figures, steps, operations, components, members, or combinations thereof.

[0046] When a given element is referred to as being "connected to", "coupled to", "supported by" or "in contact with" another element, it is to be understood that it may be directly or indirectly connected to, coupled to, supported by, or in contact with the other element. When a given element is indirectly connected to, coupled to, supported by, or in contact with another element, it is to be understood that it may be connected to, coupled to, supported by, or in contact with the other element through a third element.

[0047] It will also be understood that when an element is referred to as being "on" another element, it may be directly on the other element or intervening elements may also be present.

[0048] A refrigerator according to an embodiment of the disclosure may include a cabinet.

[0049] The "cabinet" may include an inner case, an outer case positioned outside the inner case, and an insulation provided between the inner case and the outer case.

[0050] The "inner case" may include at least one of a case, a plate, a panel, or a liner forming a storage compartment. The inner case may be formed as one body, or may be formed by assembling a plurality of plates together. The "outer case" may form an appearance of the cabinet, and be coupled to an outer side of the inner case such that the insulation is positioned between the inner case and the outer case.

[0051] The "insulation" may insulate an inside of the storage compartment from an outside of the storage compartment to maintain inside temperature of the storage compartment at appropriate temperature without being influenced by an external environment of the storage compartment. According to an embodiment of the disclosure, the insulation may include a foaming insulation. The foaming insulation may be molded by fixing the inner case and the outer case with jigs, etc., and then injecting and foaming urethane foam as a mixture of polyurethane and a foaming agent between the inner case and the outer case.

[0052] According to an embodiment of the disclosure, the insulation may include a vacuum insulation in addition to a foaming insulation, or may be configured only with a vacuum insulation instead of a forming insulation. The vacuum insulation may include a core material and a cladding material accommodating the core material and sealing the inside with vacuum or pressure close to vacuum. However, the insulation is not limited to the above-mentioned foaming insulation or vacuum insulation, and may include various materials capable of being used for insulation.

[0053] The "storage compartment" may include a space defined by the inner case. The storage compartment may further include the inner case defining the space corresponding to the storage compartment. The storage compartment may store a variety of items, such as food, medicines, cosmetics, and the like, and the storage compartment may be configured to be open on at least one side for insertion and removal of the items.

[0054] The refrigerator may include one or more storage compartments. In a case in which two or more storage compartments are formed in the refrigerator, the respective storage compartments may have different purposes of use, and may be maintained at different temperatures. To this end, the respective storage compartments may be partitioned by a partition wall including an insulation.

[0055] The storage compartment may be maintained within an appropriate temperature range according to a purpose of use, and may include a "refrigerating compartment", a "freezing compartment", and a "temperature conversion compartment" according to purposes of use and/or temperature ranges. The refrigerating compartment may be maintained at an appropriate temperature to keep food refrigerating, and the freezing compartment may be maintained at an appropriate temperature to keep food frozen. The "refrigerating" may be keeping food cold without freezing the food, and for example, the refrigerating compartment may be maintained within a range of 0 degrees Celsius to 7 degrees Celsius. The "freezing" may be freezing food or keeping food frozen, and for example, the freezing compartment may be maintained within a range of −20 degrees Celsius to −1 degrees Celsius. The temperature conversion compartment may be used as either a refrigerating compartment or a freezing compartment according to or regardless of a user's selection.

[0056] The storage compartment may also be referred to by various terms, such as "vegetable compartment", "freshness compartment", "cooling compartment", and "ice-making compartment", in addition to "refrigerating compartment", "freezing compartment", and "temperature conversion compartment", and the terms, such as "refrigerating compartment", "freezing compartment", "temperature conversion compartment", etc., as used below are to be understood as representing storage compartments having the corresponding purposes of use and the corresponding temperature ranges.

[0057] The refrigerator according to an embodiment of the disclosure may include at least one door configured to open or close the open side of the storage compartment. The respective doors may be provided to open and close one or more storage compartments, or a single door may be provided to open and close a plurality of storage compartments. The door may be rotatably or slidably mounted to the front of the cabinet.

[0058] The "door" may seal the storage compartment in a closed state. The door, like the cabinet, may include an insulation to insulate the storage compartment in a closed state.

[0059] According to an embodiment, the door may include an outer door plate forming the front surface of the door, an inner door plate forming the rear surface of the door and facing the storage compartment, an upper cap, a lower cap, and a door insulation provided therein.

[0060] A gasket may be provided on the edge of the inner door plate to seal the storage compartment by coming into close contact with the front surface of the cabinet when the door is closed. The inner door plate may include a dyke that protrudes rearward to allow a door basket for storing items to be fitted.

[0061] According to an embodiment, the door may include a door body and a front panel that is detachably coupled to the front of the door body and forming the front surface of the door. The door body may include an outer door plate forming the front surface of the door body, an inner door plate forming the rear surface of the door body and facing the storage compartment, an upper cap, a lower cap, and a door insulator provided therein.

[0062] The refrigerator may be classified as French Door Type, Side-by-side Type, Bottom Mounted Freezer (BMF), Top Mounted Freezer (TMF), or Single Door Refrigerator according to the arrangement of the doors and the storage compartments.

[0063] The refrigerator according to an embodiment of the disclosure may include a cold air supply device for supplying cold air to the storage compartment.

[0064] The "cold air supply device" may include a machine, an apparatus, an electronic device, and/or a combination system thereof, capable of generating cold air and guiding the cold air to cool the storage compartment.

[0065] According to an embodiment of the disclosure, the cold air supply device may generate cold air through a cooling cycle including compression, condensation, expansion, and evaporation processes of refrigerants. To this end, the cold air supply device may include a refrigeration cycle device **450** having a compressor, a condenser, an expander, and an evaporator to drive the refrigeration cycle. According to an embodiment of the disclosure, the cold air supply device may include a semiconductor, such as a thermoelectric element. The thermoelectric element may cool the storage compartment by heating and cooling actions through the Peltier effect.

[0066] The refrigerator according to an embodiment of the disclosure may include a machine compartment in which at least some components belonging to the cold air supply device are installed.

[0067] The "machine compartment" may be partitioned and insulated from the storage compartment to prevent heat generated by the components installed in the machine compartment from being transferred to the storage compartment. To dissipate heat from the components installed in the machine compartment, the machine compartment may communicate with outside of the cabinet.

[0068] The refrigerator according to an embodiment of the disclosure may include a dispenser provided on the door to provide water and/or ice. The dispenser may be provided on the door to allow access by the user without opening the door.

[0069] The refrigerator according to an embodiment of the disclosure may include an ice-making device that produces ice. The ice-making device may include an ice-making tray that stores water, an ice-moving device that separates ice from the ice-making tray, and an ice-bucket that stores ice produced in the ice-making tray.

[0070] The refrigerator according to an embodiment of the disclosure may include a controller for controlling the refrigerator.

[0071] The "controller" may include a memory for storing and/or recording data and/or programs for controlling the refrigerator, and a processor for outputting control signals for controlling the cold air supply device, etc., in accordance with the programs and/or data stored in the memory.

[0072] The memory may store or record various information, data, instructions, programs, and the like necessary for operation of the refrigerator. The memory may store temporary data generated while generating control signals for controlling components included in the refrigerator. The memory may include at least one of a volatile memory or a non-volatile memory, or a combination thereof.

[0073] The processor may control the overall operation of the refrigerator. The processor may control the components of the refrigerator by executing programs stored in memory. The processor may include a separate neural processing unit (NPU) that performs an artificial intelligence (AI) model operation. In addition, the processor may include a central processing unit (CPU), a graphics processor (GPU), and the like. The processor may generate a control signal to control the operation of the cold air supply device. For example, the processor may receive temperature information of the storage compartment from a temperature sensor and generate a cooling control signal to control an operation of the cold air supply device based on the temperature information of the storage compartment.

[0074] Furthermore, the processor may process a user input of a user interface and control an operation of the user interface in accordance with the programs and/or data memorized/stored in the memory. The user interface may be provided with an input interface and an output interface. The processor may receive the user input from the user interface. In addition, the processor may transmit a display control signal and image data for displaying an image on the user interface to the user interface in response to the user input.

[0075] The processor and memory may be provided integrally or may be provided separately. The processor may include one or more processors. For example, the processor may include a main processor and at least one sub-processor. The memory may include one or more memories.

[0076] The refrigerator according to an embodiment of the disclosure may include a processor and a memory for controlling all of the components included in the refrigerator, and may include a plurality of processors and a plurality of memories for individually controlling the components of the refrigerator. For example, the refrigerator may include a processor and a memory for controlling the operation of the cold air supply device in accordance with to an output of the temperature sensor. In addition, the refrigerator may be separately provided with a processor and a memory for controlling the operation of the user interface in accordance with the user input.

[0077] A communication module may communicate with external devices, such as servers, mobile devices, and other home appliances via a nearby access point (AP). The AP may connect a local area network (LAN) to which a refrigerator or a user device is connected to a wide area network (WAN) to which a server is connected. The refrigerator or the user device may be connected to the server via the WAN.

[0078] The input interface may include keys, a touch screen, a microphone, and the like. The input interface may receive the user input and pass the received user input to the processor.

[0079] The output interface may include a display, a speaker, and the like. The output interface may output various notifications, messages, information, and the like generated by the processor.

[0080] FIG. **1** is a diagram illustrating communication among a home appliance, a server, a user terminal, and the like.

[0081] A home appliance **1000** may include a communication module capable of communicating with another home appliance, a user device **2000**, or a server **3000**, a user interface that receives a user input or outputs information to a user, at least one processor that controls an operation of the home appliance **1000**, and at least one memory that stores a program for controlling the operation of the home appliance **1000**.

[0082] The home appliance **1000** may be at least one of various types of home appliances. For example, as shown in the accompanying drawings, the home appliance **1000** may include at least one of a refrigerator **1**, a dishwasher **1200**, an electric range **1300**, an electric oven **1400**, an air conditioner **1500**, a clothes treating apparatus **1600**, a washing machine **1700**, a dryer **1800**, or a microwave oven **1900**, but is not limited thereto. For example, the home appliance **1000** may include various types of appliances not shown in the drawings, such as a cleaning robot, a vacuum cleaner, a television, and the like. Furthermore, the aforementioned home appliances are by way of example only, and in addition to the aforementioned home appliances, other appliances connected to other home appliance, the user device **2000**, or the server **3000** to perform operations described below may be included in the home appliance **1000** according to an embodiment.

[0083] The server **3000** may include a communication module communicating with another server, the home appliance **1000**, or the user device **2000**, at least one processor that processes data received from another server, the home appliance **1000**, or the user device **2000**, and at least one memory that stores programs for processing data or processed data. The server **3000** may be implemented as a variety of computing devices, such as a workstation, a cloud, a data drive, a data station, and the like. The server **3000** may be implemented as one or more server physically or logically separated based on a function, detailed configuration of function, or data, and may transmit and receive data through communication between servers and process the transmitted and received data.

[0084] The server **3000** may perform functions, such as managing a user account, registering the home appliance **1000** in association with the user account, managing or controlling the registered home appliance **1000**, and the like. For example, a user may access the server **3000** via the user device **2000** and may create a user account. The user account may be identified by an identifier (ID) and a password set by the user. The server **3000** may register the home appliance **1000** with the user account according to a predetermined procedure. For example, the server **3000** may link identification information of the home appliance **1000** (e.g., a serial number or MAC address) to the user account to register, manage, and control the home appliance **1000**. The user device **2000** may include a communication module capable of communicating with the home appliance **1000** or the server **3000**, a user interface that receives a user input or outputs information to a user, at least one processor that controls an operation of the user device **2000**, and at least one memory that stores a program for controlling the operation of the user device **2000**.

[0085] The user device **2000** may be carried by a user, or placed in a user's home or office, or the like. The user device **2000** may include a personal computer (PC), a terminal, a portable telephone, a smartphone, a handheld device, a wearable device, and the like, but is not limited thereto.

[0086] The memory of the user device **2000** may store a program for controlling the home appliance **1000**, i.e. an application. The application may be sold installed on the user device **2000**, or may be downloaded from an external server for installation.

[0087] By running the application installed on the user device **2000** by a user, the user may access the server **3000**, create a user account, and communicate with the server **3000** based on the login user account to register the home appliance **1000**.

[0088] For example, by operating the home appliance **1000** to allow the home appliance **1000** to access the server **3000** according to a procedure guided by the application installed on the user device **2000**, the server **3000** may register the home appliance **1000** with the user account by assigning the identification information (e.g., a serial number or a MAC address) of the home appliance **1000** to the corresponding user account.

[0089] A user may control the home appliance **1000** using the application installed on the user device **2000**. For example, by logging into a user account with the application installed on the user device **2000**, the home appliance **1000** registered in the user account appears, and by inputting a control command for the home appliance **1000**, the control command may be delivered to the home appliance **1000** via the server **3000**.

[0090] A network may include both a wired network and a wireless network. The wired network may include a cable network or a telephone network, and the wireless network may include any networks transmitting and receiving a signal via radio waves. The wired network and the wireless network may be interconnected.

[0091] The network may include a wide area network (WAN), such as the Internet, a local area network (LAN) formed around an access point (AP), and a short-range wireless network that does not use an AP. The short-range wireless network may include Bluetooth (IEEE 802.15.1), Zigbee (IEEE 802.15.4), Wi-Fi Direct, near field communication (NFC), and Z-Wave, but is not limited thereto.

[0092] The AP may connect the home appliance **1000** or the user device **2000** to a WAN connected to the server **3000**. The home appliance **1000** or the user device **2000** may be connected to the server **3000** via a WAN.

[0093] The AP may communicate with the home appliance **1000** or the user device **2000** using wireless communication, such as Wi-Fi (IEEE 802.11), Bluetooth (IEEE 802.15.1), Zigbee (IEEE 802.15.4), and the like, and access a WAN using wired communication, but is not limited thereto.

[0094] According to various embodiments, the home appliance **1000** may be directly connected to the user device **2000** or the server **3000** without going through an AP.

[0095] The home appliance **1000** may be connected to the user device **2000** or the server **3000** via a long-range wireless network or a short-range wireless network.

[0096] For example, the home appliance **1000** may be connected to the user device **2000** via a short-range wireless network (e.g., Wi-Fi Direct).

[0097] In another example, the home appliance **1000** may be connected to the user device **2000** or the server **3000** via a WAN using a long-range wireless network (e.g., a cellular communication module).

[0098] In still another example, the home appliance **1000** may access a WAN using wired communication, and may be connected to the user device **2000** or the server **3000** via a WAN.

[0099] When accessing a WAN using wired communication, the home appliance **1000** may also act as an AP. Accordingly, the home appliance **1000** may connect another home appliance to a WAN to which the server **3000** is connected. In addition, another home appliance may connect the home appliance **1000** to the WAN to which the server **3000** is connected.

[0100] The home appliance **1000** may transmit information about an operation or state to other home appliances, the user device **2000**, or the server **3000** via the network. For example, the home appliance **1000** may transmit information about an operation or state to other home appliances, the user device **2000** or the server **3000** upon receiving a request from the server **3000**, in response to an event in the home appliance **1000**, or periodically or in real time. Upon receiving the information about the operation or state from the home appliance **1000**, the server **3000** may update the stored information about the operation or state of the home appliance **1000** and transmit the updated information about the operation and state of the home appliance **1000** to the user device **2000** via the network. Here, updating the information may include various operations in which existing information is changed, such as adding new information to the existing information, replacing the existing information with new information, and the like.

[0101] The home appliance **1000** may obtain various information from other home appliances, the user device **2000**, or the server **3000**, and may provide the obtained information to a user. For example, the home appliance **1000** may obtain information related to a function of the home appliance **1000** (e.g., recipes, washing instructions, etc.) from the server **3000** and various environmental information (e.g., weather, temperature, humidity, etc.), and may output the obtained information via a user interface.

[0102] The home appliance **1000** may operate in accordance with a control command received from other home appliances, the user device **2000**, or the server **3000**. For example, the home appliance **1000** may operate in accordance with a control command received from the server **3000**, based on a prior authorization obtained from a user to operate in accordance with the control command of the server **3000** even without a user input. Here, the control command received from the server **3000** may include a control command input by the user via the user device **2000** or a control command based on preset conditions, but is not limited thereto.

[0103] The user device **2000** may transmit information about a user to the home appliance **1000** or the server **3000** via the communication module. For example, the user device **2000** may transmit information about a user's location, a user's health condition (i.e., state), a user's preference, a user's schedule, and the like to the server **3000**. The user device **2000** may transmit information about the user to the server **3000** based on the user's prior authorization.

[0104] The home appliance **1000**, the user device **2000**, or the server **3000** may use techniques, such as artificial intelligence (AI) to determine a control command. For example, the server **3000** may receive information about an operation or a state of the home appliance **1000** or information about a user of the user device **2000**, process the received information using techniques, such as AI, and transmit a processing result or a control command to the home appliance **1000** or the user device **2000** based on the processing result.

[0105] Hereinafter, various embodiments of the refrigerator **1** among the above-described home appliance **1000** will be described in detail with reference to the accompanying drawings.

[0106] FIG. **2** illustrates a refrigerator according to an embodiment of the disclosure. FIG. **3** is a view illustrating a state in which doors of a refrigerator are opened according to an embodiment of the disclosure. FIG. **4** is a view illustrating an upper part of a storage compartment of a refrigerator, viewed from below, according to an embodiment of the disclosure. FIG. **5** is a schematic side cross-sectional view of a refrigerator according to an embodiment of the disclosure. FIG. **6** is a cross-sectional view along I-I' of FIG. **3**.

[0107] Referring to FIG. **2** to FIG. **6**, the refrigerator **1** may include a main body **100**, storage compartments **11**, **12**, and **13** formed inside the main body **100**, and doors **21**, **22**, **23**, and **24** for opening and closing the storage compartments **11**, **12**, and **13**.

[0108] The main body **100** may include an inner case, an outer case coupled to the outer side of the inner case, and an insulation **190** provided between the inner case and the outer case (see FIG. **6**). The inner case may form the storage compartments **11**, **12**, and **13**, and the outer case may form the exterior of the main body **100**.

[0109] In another aspect, the main body **100** may include an upper wall **110**, a lower wall **120**, a left wall **130**, a right wall **140**, and a rear wall **150**. The upper wall **110**, the lower wall **120**, the left wall **130**, the right wall **140**, and the rear wall **150** may form an upper side, a lower side, a left side, a right side, and a rear side of the main body **100**, respectively.

[0110] Each of the upper wall **110**, lower wall **120**, left wall **130**, right wall **140**, and rear wall **150** may be constructed with the inner case, the outer case, and the insulation **190**. For example, an upper side of the upper wall **110** may be formed by the outer case, a lower side of the upper wall **110** may be formed by the inner case, and the insulation **190** may be provided inside the upper wall **110**.

[0111] The storage compartments **11**, **12**, and **13** may accommodate items. The storage compartments **11**, **12**, and **13** may be formed so that the front is open to allow items to be put in or taken out. The main body **100** may include a horizontal partition wall **160** separating the first storage compartment **11** from the second storage compartment **12** and the third storage compartment **13**, and a vertical partition wall **161** separating the second storage compartment **12** from the third storage compartment **13**. The first storage compartment **11** may be formed in an upper part of the main body **100**, and the second storage compartment **12** and the third storage compartment **13** may be formed in a lower part of the main body **100**. The first storage compartment **11** may be a refrigerating compartment. The second storage compartment **12** may be a freezing compartment. The third storage compartment **13** may be a temperature conversion compartment.

[0112] The first storage compartment **11** may be maintained at a first set temperature, the second storage compartment **12** may be maintained at a second set temperature, and the third storage compartment **13** may be maintained at a third set temperature.

[0113] The second set temperature may be set lower than the first set temperature and the third set temperature. The second set temperature, the first set temperature, and the third set temperature may be set by a user.

[0114] The doors **21**, **22**, **23**, and **24** may open and close the storage compartments **11**, **12**, and **13**. The first door **21** and the second door **22** may open and close the first storage compartment **11**, the third door **23** may open and close the second storage compartment **12**, and the fourth door **24** may open and close the third storage compartment **13**. The doors **21**, **22**, **23**, and **24** may be rotatably coupled to the main body **100**.

[0115] The doors **21**, **22**, **23**, and **24** may be rotatably coupled to the main body **100** by hinges. For example, the first door **21** and the second door **22** may be rotatably coupled to the main body **100** by a hinge **31** disposed on the upper part of the main body **100** and a hinge disposed in the middle of the main body **100**, respectively. The hinge **31** may include a hinge pin that protrudes vertically to form a rotation axis of the door. The hinge **31** may be covered by a top cover **300** provided to cover an upper front part of the main body **100**.

[0116] A rotation bar **40** may be arranged on one of the first door **21** and the second door **22** to cover a gap between the first door **21** and the second door **22** when the first door **21** and the second door **22** are closed. The rotation bar **40** may be rotatably disposed on one of the first door **21** and the second door **22**. The rotation bar **40** may have a rod shape elongated in a vertical direction. The rotation bar **40** may also be referred to as a pillar, a mullion, and the like.

[0117] A guide protrusion **46** may be disposed at an upper end of the rotation bar **40**, and a rotating guide **119** that guides the rotation of the guide protrusion **46** may be disposed at the upper part of the main body **100**.

[0118] The doors **21**, **22**, **23**, and **24** may include a gasket **51**. The gasket **51** may make close contact with a front side of the main body **100** when the doors **21**, **22**, **23**, and **24** are closed. The doors **21**, **22**, **23**, and **24** may each include a dyke **52** that protrudes rearward. A door shelf **53** capable of storing items may be mounted on the dyke **52**. The rotation bar **40** may be rotatably installed on the dyke **52**.

[0119] Although the number and arrangement of the storage compartments and the number and arrangement of the doors have been described above, the number and arrangement of the storage compartments and the number and arrangement of the doors of the refrigerator according to an embodiment of the disclosure are not limited.

[0120] The refrigerator **1** may include a thermoelectric cooler **400** to cool the storage compartment **11**.

[0121] The thermoelectric cooler **400** may be disposed above the storage compartment **11** to cool the storage compartment **11**. That is, the thermoelectric cooler **400** may be disposed on the upper wall **110** of the main body **100**.

[0122] The thermoelectric cooler **400** may include a thermoelectric element **530**. The thermoelectric element **530** may be a semiconductor device that converts thermal energy into electrical energy and converts electrical energy into thermal energy using the thermoelectric effect. The thermoelectric element **530** may be referred to by various terms, such as a semiconductor thermoelectric element and a Peltier element.

[0123] The thermoelectric element **530** includes a heating portion **531** and a cooling portion **532**. When current is applied to the thermoelectric element **530**, heating may occur in the heating portion **531** and heat absorption may occur in the cooling portion **532**. The thermoelectric element **530** may have a thin hexahedral shape. The heating portion **531** may be formed on one side of the thermoelectric element **530**, and the cooling portion **532** may be formed on the opposite side.

[0124] The thermoelectric element **530** may be disposed on the upper wall **110** such that the heating portion **531** may be positioned above the thermoelectric element **530** and the cooling portion **532** may be positioned below the thermoelectric element **530**. That is, the heating portion **531** may face the outside of the main body **100** and the cooling portion **532** may face the inside of the storage compartment **11**. Accordingly, the air that has been warmed by heat exchange with the heating portion **531** may be discharged to the outside of the main body **100**, and the air that has been cooled by heat exchange with the cooling portion **532** may be supplied to the storage compartment **11**.

[0125] The thermoelectric cooler **400** may include a heat sink **520** that contacts the heating portion **531** to allow heat exchange between the heating portion **531** and the outside air of the main body **100** to be efficiently performed.

[0126] The heat sink **520** may be located on the outside of the main body **100**. The heat sink **520** may contact the heating portion **531** to absorb heat from the heating portion **531** and dissipate heat to the outside of the main body **100**. The heat sink **520** may be referred to by various terms, such as a hot sink, a heatsink, and a hot heat sink.

[0127] The heat sink **520** may be formed of a metal material having a high thermal conductivity. For example, the heat sink **520** may be formed of aluminum or copper.

[0128] The heat sink **520** may include a heat sink base **521** that contacts the heating portion **531**, and a plurality of heat dissipation fins **525** that protrude from the heat sink base **521** to expand a heat transfer area. The plurality of heat dissipation fins **525** may protrude upward from the heat sink base **521**.

[0129] The thermoelectric cooler **400** may include a cooling sink **570** that contacts the cooling portion **532** to allow heat exchange between the cooling portion **532** and the air in the storage compartment **11** to be efficiently performed.

[0130] The cooling sink **570** may be located inside the storage compartment **11**. The cooling sink **570** may absorb heat from the storage compartment **11** and transfer to the cooling portion **532**. Accordingly, the storage compartment **11** may be cooled. The cooling sink **570** may be referred to by various terms, such as a cold sink, a cold heat sink, and a cooling heat sink.

[0131] The cooling sink **570** may be formed of a metal material having a high thermal conductivity. For example, the cooling sink **570** may be formed of aluminum or copper.

[0132] The cooling sink **570** may include a cooling sink base **571** that contacts the cooling portion **532**, and a plurality of cooling fins **575** that protrude from the cooling sink base **571** to expand a heat transfer area. The plurality of cooling fins **575** may protrude downward from the cooling sink base **571**. The cooling sink base **571** and the plurality of cooling fins **575** may be formed integrally.

[0133] The thermoelectric cooler **400** may include a heat dissipation fan **600** that causes air to flow to allow heat exchange between the heat sink **520** and the outside air of the main body **100** to be efficiently performed.

[0134] The heat dissipation fan **600** may blow air toward the heat sink **520**. The heat dissipation fan **600** may be positioned in a horizontal direction of the heat sink **520**. The heat dissipation fan **600** may be disposed on the outside of the main body **100**. The heat dissipation fan **600** may be disposed on the upper side of the upper wall **110**.

[0135] The heat dissipation fan **600** may be a centrifugal fan that draws in air in an axial direction and discharges the air in a radial direction. The centrifugal fan may include a blower fan. A rotation axis **610** of the heat dissipation fan **600** may be disposed vertically on the upper side of the upper wall **110**.

[0136] The thermoelectric cooler **400** may include a heat dissipation duct **700** to guide air flowing by the heat dissipation fan **600**. The heat dissipation duct **700** may guide the outside air of the main body **100** to exchange heat with the heat sink **520**. Air that has exchanged heat with the heat sink **520** may be discharged to the outside of the main body **100** through the heat dissipation duct **700**.

[0137] The heat dissipation duct **700** may draw in the air from an external space above the main body **100**. The heat dissipation duct **700** may discharge the air that has exchanged heat with the heat sink **520** to the external space above the main body **100**. The heat dissipation fan **600** may be located inside the heat dissipation duct **700**. The heat sink **520** may be located inside the heat dissipation duct **700**. The heat dissipation duct **700** may be disposed on the upper side of the upper wall **110**.

[0138] The heat dissipation duct **700** may include an outside air inlet **751** for drawing the outside air of the main body **100** into the heat dissipation duct **700**, and an outside air outlet **782** for discharging air that has exchanged heat with the heat sink **520** to the outside of the main body **100**.

[0139] The thermoelectric cooler **400** may include a cooling fan **800** that causes air to flow to allow heat exchange between the cooling sink **570** and the air in the storage compartment **11** to be efficiently performed.

[0140] The cooling fan **800** may blow air toward the cooling sink **570**. The cooling fan **800** may be positioned in a horizontal direction of the cooling sink **570**. The cooling fan **800** may be disposed inside the storage compartment **11**. The cooling fan **800** may be disposed on the lower side of the upper wall **110**.

[0141] The cooling fan **800** may be a centrifugal fan that draws in air in the axial direction and discharges the air in a radial direction. A rotation axis **810** of the cooling fan **800** may be disposed vertically on the lower side of the upper wall **110**.

[0142] The thermoelectric cooler **400** may include the cooling duct **900** to guide air flowing by the cooling fan **800**. The cooling fan **800** may guide the air in the storage compartment **11** to exchange heat with the cooling sink **570**. Air that has exchanged heat with the cooling sink **570** may be discharged back into the storage compartment **11** through the cooling duct **900**.

[0143] The cooling fan **800** may be located inside the cooling duct **900**. The cooling sink **570** may be located inside the cooling duct **900**. The cooling duct **900** may be disposed on the lower side of the upper wall **110**.

[0144] The cooling duct **900** may include an internal air inlet **991** for drawing air inside the storage compartment **11** into the cooling duct **900**, and an internal air outlet **992** for discharging the air that has exchanged heat with the cooling sink **570** into the storage compartment **11**.

[0145] Referring to FIG. **5**, the refrigerator **1** may include the refrigeration cycle device **450** to cool the storage compartment through a refrigeration cycle. The refrigeration cycle device **450** may include a compressor **2**, a condenser (not shown), an expansion device (not shown), and an evaporator **3**. The evaporator **3** may be disposed at the rear of the storage compartments **12** and **13**.

[0146] According to various embodiments, the evaporator may not be provided at the rear of the first storage compartment **11**. That is, the refrigerator **1** according to an embodiment may include only one evaporator **3**, and the evaporator **3** may be disposed at the rear of the second storage compartment **12**. The evaporator **3** may also be disposed at a lower part based on the horizontal partition wall **160**.

[0147] The refrigerator **1** may include a defrost sensor **111** for measuring a temperature of the evaporator **3**.

[0148] The defrost sensor **111** may measure the temperature of the evaporator **3**. Measuring the temperature of the evaporator **3** may include measuring a temperature of the air around the evaporator **3** and measuring a temperature of the evaporator **3** itself.

[0149] The defrost sensor **111** may be provided on the evaporator **3** or may be provided in evaporator ducts **60** and **70**.

[0150] The refrigerator **1** may include the evaporator ducts **60** and **70** to guide cold air generated in the evaporator **3**. The first evaporator duct **60** may be disposed at the rear of the second storage compartment **12** and the third storage compartment **13**. The second evaporator duct **70** may be disposed at the rear of the first storage compartment **11**.

[0151] The cold air generated in the evaporator **3** may be drawn into the first evaporator duct **60** by the evaporator fan **80**. The cold air drawn into the first evaporator duct **60** may be discharged to the second storage compartment **12** or the third storage compartment **13** through a cold air outlet (not shown) formed at the front. In addition, the cold air drawn into the first evaporator duct **60** may be guided to an internal flow path **78** of the second evaporator duct **70**. The first evaporator duct **60** may be provided with a damper **61** to control the supply of the cold air in the first evaporator duct **60** to the second evaporator duct **70**. A connection duct **90** may be disposed between the first evaporator duct **60** and the second evaporator duct **70** to connect the first evaporator duct **60** and the second evaporator duct **70**.

[0152] The internal flow path **78** of the second evaporator duct **70** may guide the cold air generated in the evaporator **3** to the first storage compartment **11**.

[0153] The damper **61** may open or close the internal flow path **78**.

[0154] When the internal flow path **78** is opened by the damper **61**, the cold air generated in the evaporator **3** may be guided to the first storage compartment **11**.

[0155] When the internal flow path **78** is closed by the damper **61**, the cold air generated in the evaporator **3** may be blocked by the damper **61** and may not be guided to the first storage compartment **11**.

[0156] The cold air flowing into the internal flow path **78** of the second evaporator duct **70** may be supplied to the first storage compartment **11** through a cold air outlet **72** formed on the front side of the second evaporator duct **70**.

[0157] However, unlike the above embodiment, the cold air generated in the evaporator **3** may be supplied directly to the second evaporator duct **70** without passing through the first evaporator duct **60**. In addition, a separate evaporator **3** for supplying cold air to the second evaporator duct **70** may be provided at the rear of the first storage compartment **11**.

[0158] As such, the refrigerator **1** according to an embodiment of the disclosure may include the thermoelectric cooler **400** and the refrigeration cycle device **450** for cooling the storage compartment. Accordingly, the storage compartment may be cooled by using at least one of the thermoelectric cooler **400** or the refrigeration cycle device **450**. For example, the storage compartment may be cooled by supplying only the cold air generated by the refrigeration cycle device **450**, or by supplying only the cold air generated by the thermoelectric cooler **400**. In addition, the storage compartment may be cooled by supplying the cold air generated by the thermoelectric cooler **400** together with the cold air generated by the refrigeration cycle device **450**.

[0159] The refrigerator **1** may supply cold air to the storage compartment **11** according to external and internal conditions. For example, in a case where an outside temperature of the refrigerator **1** is higher or lower than a preset temperature range, cooling by the refrigeration cycle device **450** is more efficient than cooling by the thermoelectric cooler **400**. Accordingly, in a case where the outside temperature of the refrigerator **1** is higher or lower than the preset temperature range, the storage compartment **11** may be cooled only by the cold air generated by the refrigeration cycle device **450**.

[0160] In a case where the outside temperature of the refrigerator **1** is within the preset temperature range and the storage compartment **11** is overloaded or the storage compartment **11** requires to be rapidly cooled, the cold air generated by the refrigeration cycle device **450** and the cold air generated by the thermoelectric cooler **400** may be supplied to the storage compartment **11** simultaneously to cool the storage compartment **11** rapidly.

[0161] Meanwhile, although it has been described that the thermoelectric cooler **400** is disposed on the upper wall **110** of the main body **100**, the location of the thermoelectric cooler **400** is not limited thereto.

[0162] According to various embodiments, the thermoelectric cooler **400** may be provided on at least one of the upper wall **110**, the lower wall **120**, the left wall **130**, the right wall **140**, or the rear wall **150**.

[0163] FIG. **7** is an exploded view of a thermoelectric cooler according to an embodiment of the disclosure.

[0164] Referring to FIG. **7**, the thermoelectric cooler **400** may include a thermoelectric module **500**.

[0165] The above-described thermoelectric element **530**, the heat sink **520**, and the cooling sink **570** may be integrally assembled as the thermoelectric module **500**. That is, the thermoelectric module **500** may include the thermoelectric element **530**, the heat sink **520**, the cooling sink **570**, and a module plate **550**.

[0166] The module plate **550** may serve as a frame of the thermoelectric module **500**. The module plate **550** may be formed of a resin material having a low thermal conductivity. The module plate **550** may maintain a gap between the heat sink **520** and the cooling sink **570**, and may support the heat sink **520** and the cooling sink **570**. The module plate **550** may be formed integrally with a fan case **650** to be described below. However, the module plate **550** may be provided separately from the fan case **650**.

[0167] The module plate **550** may include a heat sink support **552** supporting the heat sink **520**.

[0168] The module plate **550** may include a module plate opening **551**. The thermoelectric element **530** may be disposed inside the module plate opening **551**. A vertical length of the module plate opening **551** may be larger than that of the thermoelectric element **530**, and the thermoelectric element **530** may be disposed at an upper end of the module plate opening **551**. In general, the heat generation amount of the thermoelectric element **530** is higher than the heat absorption amount, and it is advantageous for heat dissipation of the heating portion **531** that the thermoelectric element **530** is located at the upper end of the module plate opening **551**. Accordingly, the thermoelectric element **530** may be disposed at the upper end of the module plate opening **551**.

[0169] As such, because the thermoelectric element **530** is disposed at the upper end of the module plate opening **551**, the cooling sink **570** may include a cooling conductive portion **574** protruding from the cooling sink base **571** to contact the cooling portion **532** of the thermoelectric element **530**.

[0170] The thermoelectric module **500** may include an element insulation **540** to insulate the module plate **550** and the thermoelectric element **530**. The element insulation **540** may be disposed in the module plate opening **551** to prevent sides of the thermoelectric element **530** from contacting the module plate **550**. The element insulation **540** may include an element insulation opening **541**, and the thermoelectric element **530** may be accommodated in the element insulation opening **541**.

[0171] The thermoelectric module **500** may include a sink insulation **580** between the module plate **550** and the cooling sink **570**. The sink insulation **580** may prevent heat from being transferred between the heat sink **520** and the cooling sink **570** through the module plate **550**. The sink insulation **580** may include a sink insulation opening **581**. However, the sink insulation **580** may be omitted. In this case, the heat sink **520** may be supported on an upper side of the module plate **550** and the cooling sink **570** may be supported on a lower side of the module plate **550**.

[0172] The thermoelectric cooler **400** may include the fan case **650** on which the heat dissipation fan **600** is installed, and the fan case **650** may guide air flowing by the heat dissipation fan **600**.

[0173] The fan case **650** may be formed integrally with the module plate **550** or may be provided separately.

[0174] The fan case **650** may include a case bottom **660** on which the heat dissipation fan **600** is rotatably installed, and a case scroll portion **670** extending upward from an edge of the case bottom **660** to guide air blown from the heat dissipation fan **600** toward the heat sink **520**. The heat dissipation fan **600** may be a centrifugal fan, and may be installed on the case bottom **660** to allow the rotation axis **610** to be perpendicular to the case bottom **660**. In addition, the heat sink **520** may be positioned in a radial direction of the heat dissipation fan **600**. With the structure described above, an overall vertical length of the thermoelectric cooler **400** may be compact.

[0175] The case scroll portion **670** may be formed to surround the heat dissipation fan **600**. The case scroll portion **670** may have a scroll portion opening **673** open toward the heat sink **520**. The case scroll portion **670** may include a downstream end **671** along a rotational direction R of the heat dissipation fan **600** and an upstream end **672** along the rotational direction R.

[0176] The fan case **650** may include a case guide **680** to guide air flowing from the heat dissipation fan **600** to the vicinity of the downstream end **671** of the case scroll portion **670**.

[0177] The heat sink **520** may include a plurality of heat dissipation fins **525**. The plurality of heat dissipation fins **525** may protrude from an upper side **522** of the heat sink base **521**. The plurality of heat dissipation fins **525** may protrude in a direction perpendicular to the upper side **522** of the heat sink base **521**.

[0178] A plurality of heat dissipation channels may be formed between the plurality of heat dissipation fins **525**.

[0179] The heat dissipation fan **600** may blow air toward the heat sink **520**, and the air flowing by the heat dissipation fan **600** may pass through the heat dissipation channels and exchange heat with the plurality of heat dissipation fins **525**.

[0180] The cooling sink **570** may include a plurality of cooling fins **575**. The plurality of cooling fins **575** may be formed to extend in a direction parallel to a lower side of the cooling sink base **571**.

[0181] A plurality of cooling channels may be formed between the plurality of cooling fins **575**.

[0182] The air flowing by the cooling fan **800** may pass through the cooling channels and exchange heat with the plurality of cooling fins **575**.

[0183] FIG. **8** is a block diagram illustrating an example configuration of a refrigerator according to an embodiment of the disclosure.

[0184] Referring to FIG. **8**, the refrigerator **1** according to an embodiment may include a current sensor **115**, a user interface **200**, a communication interface **250**, the first cooler **400**, the second cooler **450**, and a controller **350**. The controller **350** may include at least one processor **351** and memory **352**.

[0185] The current sensor **115** may detect a current flowing through the thermoelectric element **530**. The current sensor **115** may transmit information about the current flowing through the thermoelectric element **530** to the controller **350**.

[0186] The refrigerator **1** may include the user interface **200**.

[0187] The user interface **200** may convert sensory information received from a user into an electrical signal.

[0188] The user interface **200** may include a power button, an operation button, a menu selection button, refrigeration/freezing setting button, a rapid cooling setting button, and the like. For example, the user interface **200** may include a tact switch, a push switch, a slide switch, a toggle switch, a micro switch, a touch switch, a touch pad, a touch screen, a jog dial, and/or a microphone.

[0189] The user interface **200** may visually or audibly transmit information related to the operation of the refrigerator **1** to a user. Information about the operation of the refrigerator **1** may be output via a screen, an indicator, or a voice. For example, the user interface **200** may include a liquid crystal display (LCD) panel, a light emitting diode (LED) panel, or a speaker.

[0190] The refrigerator **1** may include the communication interface **250** for wired and/or wireless communication with an external device.

[0191] The communication interface **250** may include at least one of a short-range wireless communication module or a long-range wireless communication module.

[0192] The communication interface **250** may transmit data to an external device (e.g., a server, a user device, a temperature probe) or receive data from the external device. For the communication, the communication interface **250** may establish a direct (e.g., wired) communication channel or a wireless communication channel between the external devices, and support the performance of the communication through the established communication channel. According to an embodiment, the communication interface **250** may include a wireless communication module (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module (e.g., a local area network (LAN) communication module, or a power line communication module). Among these communication modules, the corresponding communication module may communicate with an external device through a first network (e.g., a short-range wireless communication network such as Bluetooth, wireless fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or a second network (e.g., a long-range wireless communication network such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network WAN)). These various types of communication modules may be integrated as a single component (e.g., a single chip) or implemented as a plurality of separate components (e.g., multiple chips).

[0193] The short-range wireless communication module may include a Bluetooth communication module, a Bluetooth Low Energy (BLE) communication module, a near field communication module, a WLAN (Wi-Fi) communication module, and a Zigbee communication module, an IrDA communication module, a Wi-Fi Direct (WFD) communication module, an ultrawideband (UWB) communication module, an Ant+ communication module, a microwave (uWave) communication module, etc., but is not limited thereto.

[0194] The long-range wireless communication module may include a communication module that performs various types of long-range wireless communication, and may include a mobile communication interface. The mobile communication interface transmits and receives radio signals with at least one of a base station, an external terminal, and a server in a mobile communication network.

[0195] According to an embodiment, the communication interface **250** may communicate with an external device through an access point (AP). The AP may connect a LAN, to which the refrigerator **1** is connected, to a WAN to which a server is connected. The refrigerator **1** may be connected to the server via the WAN.

[0196] The refrigerator **1** may receive various signals (e.g., weather information, remote instructions) from an external device (e.g., a server, a user device) through the communication interface **250**.

[0197] The refrigerator **1** may transmit various signals to the external device through the communication interface **250**.

[0198] The refrigerator **1** may include the first cooler **400** configured to cool the first storage compartment **11**. The first cooler **400** may be the thermoelectric cooler **400** described above.

[0199] The thermoelectric cooler **400** may include the thermoelectric element **530**, the heat dissipation fan **600**, and/or the cooling fan **800**. In addition, the thermoelectric cooler **400** may further include a heat absorption sensor **850** for detecting a temperature of air absorbed by the thermoelectric element **530** and a heat dissipation sensor **860** for detecting a temperature of air discharged by the thermoelectric element **530**.

[0200] When power is supplied, the thermoelectric element **530** may allow heat exchange to occur between the cooling sink **570** and the heat sink **520**. For example, the thermoelectric element **530** may convert electrical energy into heat energy to cause a heat generation process in the heating portion **531** and a heat absorption process in the cooling portion **532**.

[0201] When heat generation occurs in the heating portion **531**, air warmed by the heat sink **520** that contacts the heating portion **531** may be discharged to the outside of the main body **100**, and air cooled by the cooling sink **570** that contacts the cooling portion **532** may be supplied to the first storage compartment **11**.

[0202] The controller **350** may control the thermoelectric element **530**. Controlling the thermoelectric element **530** may include controlling the on/off of the thermoelectric element **530**. Controlling the thermoelectric element **530**

may include controlling a drive circuit that supplies power to the thermoelectric element **530**.

[0203] Driving the thermoelectric element **530** may include supplying electrical energy to the thermoelectric element **530**, i.e., supplying power to the thermoelectric element **530**. Supplying power to the thermoelectric element **530** may include applying voltage and/or current to the thermoelectric element **530**.

[0204] Driving the thermoelectric element **530** may include pulse width modulation (PWM) controlling the thermoelectric element **530**.

[0205] Turning off the thermoelectric element **530** may include not supplying electrical energy to the thermoelectric element **530**, i.e., not supplying power to the thermoelectric element **530**. Not supplying power to the thermoelectric element **530** may include not applying voltage and/or current to the thermoelectric element **530**. Not supplying power to the thermoelectric element **530** may include not PWM controlling the thermoelectric element **530**.

[0206] In the disclosure, turning off the thermoelectric element **530** may not include intermittently not supplying power to the thermoelectric element **530** according to an on/off duty ratio while PWM controlling the thermoelectric element **530**. That is, even though power is not supplied intermittently to the thermoelectric element **530** according to the on/off duty ratio while the thermoelectric element **530** is PWM controlled, the thermoelectric element **530** is still being driven.

[0207] As the thermoelectric element **530** is driven, the heat sink **520** may contact the heating portion **531** to absorb the heat of the heating portion **531** and may release the heat to the outside of the main body **100**.

[0208] As the thermoelectric element **530** is driven, the cooling sink **570** may cool the first storage compartment **11** by removing the heat from the storage compartment **11** and transferring the heat to the cooling portion **532**.

[0209] In an embodiment, in a cooling mode, the controller **350** may control the thermoelectric element **530** to maintain a temperature of the first storage compartment **11** at a set temperature of the first storage compartment **11** (hereinafter, 'first set temperature'). The set temperature of the first storage compartment **11** may be set via the user interface **200** of the refrigerator **1** or may be set remotely from an external device via the communication interface **250**.

[0210] The heat dissipation fan **600** may draw in air from outside the main body **100** and guide the drawn air to exchange heat with the heat sink **520**, and may discharge the air that has exchanged heat with the heat sink **520** back to the outside of the main body **100**.

[0211] The controller **350** may control the heat dissipation fan **600**. Controlling the heat dissipation fan **600** may include controlling a fan motor of the heat dissipation fan **600**. Controlling the heat dissipation fan **600** may include driving the heat dissipation fan **600** and turning off the heat dissipation fan **600**. Driving the heat dissipation fan **600** may include rotating the heat dissipation fan **600** at a predetermined speed. Turning off the heat dissipation fan **600** may include stopping the rotation of the heat dissipation fan **600**.

[0212] The fan motor of the heat dissipation fan **600** may include a brushless direct current (BLDC) motor whose speed may be controlled.

[0213] According to the operation of the heat dissipation fan **600**, the air that has exchanged heat with the heat sink **520** flows, thereby allowing the heat sink **520** to quickly dissipate heat. As the heat sink **520** quickly dissipates heat, the heat generation in the heating portion **531** and the heat absorption in the cooling portion **532** may occur smoothly.

[0214] The cooling fan **800** may draw in air from the storage compartment **11**, allow the drawn air to exchange heat with the cooling sink **570**, and discharge the air that has exchanged heat with the cooling sink **570** back into the storage compartment **11**.

[0215] The controller **350** may control the cooling fan **800**. Controlling the cooling fan **800** may include controlling a fan motor of the cooling fan **800**. Controlling the cooling fan **800** may include driving the cooling fan **800** and turning off the cooling fan **800**. Driving the cooling fan **800** may include rotating the cooling fan **800** at a predetermined speed. Turning off the cooling fan **800** may include stopping the rotation of the cooling fan **800**.

[0216] The fan motor of the cooling fan **800** may include a BLDC motor whose speed may be controlled.

[0217] According to the operation of the cooling fan **800**, the air that has exchanged heat with the cooling sink **570** flows, thereby rapidly cooling the inside of the storage compartment **11**. As the air that has exchanged heat with the cooling sink **570** flows, the heat generation in the heating portion **531** and the heat absorption in the cooling portion **532** may occur smoothly.

[0218] In an embodiment, the controller **350** may operate the cooling fan **800** and the heat dissipation fan **600** based on the thermoelectric element **530** being turned on. The controller **350** may turn off the cooling fan **800** and the heat dissipation fan **600** based on the thermoelectric element **530** being turned off.

[0219] In an embodiment, the controller **350** may operate the cooling fan **800** and the heat dissipation fan **600** based on the thermoelectric element **530** being turned off in a defrosting mode of the thermoelectric element **530**. The defrosting mode of the thermoelectric element **530** may be a mode in which the heat dissipation fan **600** and the cooling fan **800** are driven while the thermoelectric element **530** is not driven, in order to defrost the thermoelectric element **530**. That is, the controller **350** may turn off the thermoelectric element **530** and drive the cooling fan **800** and the heat dissipation fan **600** to defrost the thermoelectric element **530**.

[0220] As such, the refrigerator **1** according to an embodiment of the disclosure may supply the cold air generated by the first cooler **400** to the storage compartment to cool the storage compartment. Cooling the storage compartment **11** by supplying the cold air generated by the first cooler **400** to the storage compartment **11** is referred to as first cooling.

[0221] Meanwhile, the refrigerator **1** may include the second cooler **450** configured to supply cold air to the first storage compartment **11** and/or the second storage compartment **12**. The second cooler **450** may be the refrigeration cycle device **450** described above.

[0222] The second cooler **450** may include the compressor **2** and the evaporator fan **80**.

[0223] The compressor **2** may compress a refrigerant and supply the compressed refrigerant to a heat exchanger (e.g., a condenser (not shown), an expansion device (not shown), and the evaporator **3**).

[0224] The controller **350** may control a temperature of the cold air generated in the evaporator **3** by controlling the compressor **2**. For example, the controller **350** may control the compressor **2** to maintain the temperature measured by the internal sensor **112** at a predetermined target temperature.

[0225] Controlling the compressor **2** may include controlling the on/off of the compressor **2** or controlling an operating frequency of the compressor **2**.

[0226] The controller **350** may control the evaporator fan **80** to blow the cold air generated in the evaporator **3** to the first storage compartment **11** and/or the second storage compartment **12**.

[0227] As such, the refrigerator **1** according to an embodiment of the disclosure may cool the storage compartment by supplying the cold air generated by the second cooler **450** to the storage compartment. Cooling the storage compartment **11** by supplying the cold air generated by the second cooler **450** to the storage compartment **11** is referred to as second cooling.

[0228] The controller **350** may include at least one processor **351** controlling the operation of the refrigerator **1**, and at least one memory **352** storing programs and data for controlling the operation of the refrigerator **1**.

[0229] The at least one memory **352** may store data required for various embodiments. The at least one memory **352** may be implemented as a memory embedded in the refrigerator **1**, or as a memory detachable from the refrigerator **1** depending on a data storage use. For example, data for driving the refrigerator **1** may be stored in the memory embedded in the refrigerator **1**, and data for an extended function of the refrigerator **1** may be stored in the memory detachable from the refrigerator **1**. Meanwhile, the memory embedded in the refrigerator **1** may be implemented as at least one of a volatile memory (e.g., a dynamic random access memory (DRAM), static RAM (SRAM), and/or synchronous dynamic RAM (SDRAM)), a non-volatile memory (e.g., an one time programmable read only memory (OTPROM), programmable ROM (PROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), mask ROM, flash ROM, flash memory (e.g., NAND flash, NOR flash, etc.), a hard drive, or a solid state drive (SSD)). In addition, the memory detachable from the refrigerator **1** may be implemented as a memory card (e.g., a compact flash (CF), a secure digital (SD), a micro secure digital (Micro-SD), a mini secure digital (Mini-SD), an extreme digital (xD), or a multi-media card (MMC)), an external memory (e.g., universal serial bus (USB) memory) connectable to a USB port, and the like.

[0230] The at least one processor **351** may control an overall operation of the refrigerator **1**. Specifically, the at least one processor **351** may be connected to various components (the current sensor **115**, the user interface **200**, the communication interface **250**, the first cooler **450**, the second cooler **400**, etc.) of the refrigerator **1** and may control the overall operation of the refrigerator **1**. For example, the at least one processor **351** may be electrically connected to the memory **352** and may control the overall operation of the refrigerator **1**. The processor **351** may be formed as a single processor or a plurality of processors.

[0231] The processor **351** may perform various operations of the refrigerator **1** by processing at least one instruction stored in the memory **352**.

[0232] The at least one processor **351** may include at least one of a central processing unit (CPU), graphics processing unit (GPU), accelerated processing unit (APU), many integrated core (MIC), digital signal processor (DSP), neural processing unit (NPU), hardware accelerator, or machine learning accelerator. The at least one processor **351** may control one or any combination of other components of the refrigerator **1** and may perform operations related to communication or data processing. The at least one processor **351** may execute at least one program or instruction stored in the memory **352**. For example, the at least one processor **351** may execute at least one instruction stored in the memory **352** to perform a method according to at least one embodiment of the disclosure.

[0233] The refrigerator **1** according to an embodiment of the disclosure includes the first cooler **400** and the second cooler **450** for cooling the storage compartment **11**, and may cool the storage compartment **11** by using at least one of the first cooler **400** or the second cooler **450**. For example, in order to cool the storage compartment **11**, the refrigerator **1** may perform only the first cooling by the first cooler **400**, may perform only the second cooling by the second cooler **450**, or may perform the first cooling by the first cooler **400** and the second cooling by the second cooler **450** simultaneously.

[0234] In a case where the first cooling by the first cooler **400** is performed, the refrigerator **1** may perform control to increase an operating intensity of the second cooler **450** based on a failure in the first cooler **400**, which is described in detail below.

[0235] FIG. **9** is a flowchart illustrating a method for controlling a refrigerator according to an embodiment of the disclosure.

[0236] In a case where cooling is performed by the first cooler **400**, i.e., in a state where the first cooler **400** is turned on, the first cooling may not be performed smoothly due to a failure in the first cooler **400**, and the like.

[0237] The at least one processor **351** may detect whether a failure has occurred in the first cooler **400**. A failure in the first cooler **400** may be detected based on various states, as described below. Detecting whether a failure has occurred in the first cooler **400** and various operations depending on the failure may be performed by the at least one processor **351**, but may also be performed by the server **3000** communicating with the refrigerator **1**, etc., as described above. The server **3000** may detect whether a failure has occurred in the first cooler **400** and may transmit the detection result to the refrigerator **1**, the user device **2000**, or the like.

[0238] In response to detecting the failure in the first cooler **400** (Yes in operation **901**), the at least one processor **351** may increase an operating intensity of the second cooler **450** to maintain a cooling performance. That is, an operating speed of the compressor **2** and the evaporator fan **80** included in the second cooler **450** may be increased (**903**). Here, the operating speed may be an example of an operating intensity of the compressor **2** and the evaporator fan **80**.

[0239] Thereafter, the at least one processor **351** may perform control to increase or decrease a target temperature of the storage compartment (**905**).

[0240] In an embodiment, the at least one processor **351** may decrease the target temperature of the storage compartment. Cooling the storage compartment is performed only by the second cooler **450** due to the failure of the first cooler **400**, and thus considering that it may be difficult to reach an existing target temperature, the at least one processor **351** may decrease the target temperature of the storage compartment.

[0241] In the above-described embodiment, it has been described that the target temperature is reduced after increasing the operating speed of the compressor **2** and the evaporator fan **80**. However, the operating speed of the compressor **2** and the evaporator fan **80** may also be increased by decreasing the target temperature of the storage compartment without separately controlling the compressor **2** and the evaporator fan **80**.

[0242] In another embodiment, the at least one processor **351** may increase the target temperature of the storage compartment.

[0243] In order to efficiently maintain a proper temperature and proper humidity, the at least one processor **351** may increase the target temperature of the storage compartment, considering that significant power may be consumed to cool the storage compartment with only the second cooler **450**.

[0244] The at least one processor **351** may increase or decrease the target temperature of the storage compartment considering various conditions such as external situations or power consumption.

[0245] Hereinafter, various processes for detecting whether a failure has occurred in the first cooler **400** are described.

[0246] FIG. **10** is a flowchart illustrating operations of a refrigerator depending on whether a failure has occurred in a thermoelectric cooler based on a current detection result according to an embodiment of the disclosure.

[0247] As an example of detecting an occurrence of a failure in the first cooler **400**, the at least one processor **351** may detect that a failure has occurred in the first cooler **400** in a case where a current flowing through the thermoelectric element **530** is outside a reference range.

[0248] That is, in a case where the current sensor **115** detects the current flowing through the thermoelectric element **530** (**1001**), and the detected current value is determined to be outside the reference current range for a first period of time (Yes in operation **1003**), the at least one processor **351** may detect that a failure has occurred in the first cooler **400**.

[0249] Here, the first period of time may be set to an appropriate period of time for detecting whether a failure has occurred in the thermoelectric element **530**, and may be, for example, 60 seconds. The reference current range may also be an appropriate current value range for detecting whether a failure has occurred in the thermoelectric element **530**, and may be, for example, 0.5 A to 4.5 A.

[0250] That is, in a case where the current flowing through the thermoelectric element **530** is detected to be outside the range of 0.5 A to 4.5 A for 60 seconds, the at least one processor **351** may detect that a failure has occurred in the first cooler **400**.

[0251] It has been described in the above embodiment that the first cooler **400** is detected as faulty in a case where a current value remains outside the reference current range for a predetermined period of time. However, in another embodiment, a predetermined margin may be added to the reference current range, and the first cooler **400** may be detected as faulty in a case where a current value remains outside an extended range including the predetermined margin for a predetermined period of time. For example, in a case where the reference current range is 0.5 A to 4.5 A and a current value is detected to be outside an extended range of 0.4 A to 5.4 A for a predetermined period of time, the first cooler **400** is detected as faulty.

[0252] In connection with determining whether a failure has occurred in the first cooler **400**, the refrigerator **1** according to another embodiment may include a voltage sensor **116**. The voltage sensor **116** may detect a voltage applied to the thermoelectric element **530**. The voltage sensor **116** may transmit information about the voltage applied to the thermoelectric element **530** to the controller **350**.

[0253] That is, as another example of detecting an occurrence of a failure in the first cooler **400**, the at least one processor **351** may detect that a failure has occurred in the first cooler **400** in a case where the voltage applied to the thermoelectric element **530** is outside a reference range.

[0254] That is, when the voltage sensor **116** detects the voltage applied to the thermoelectric element **530** and the detected voltage value is determined to be outside a reference voltage range for the first period of time, the at least one processor **351** may detect that a failure has occurred in the first cooler **400**.

[0255] Here, the reference voltage range may be an appropriate voltage value range for detecting whether a failure has occurred in the thermoelectric element **530**.

[0256] In connection with determining whether a failure has occurred in the first cooler **400**, an artificial intelligence (AI) model may be used to determine whether a failure has occurred in the first cooler **400**. The AI model may be stored in the memory **352** of the refrigerator **1** or may be stored in a separate external device such as a server.

[0257] The at least one processor **351** may input, to the AI model, the current value detected by the current sensor **115** or the voltage value detected by the voltage sensor **116** as input data, and information about the failure of the first cooler **400** based on the corresponding current value or voltage value as output data.

[0258] The AI model may be trained based on the input data about the current value detected by the current sensor **115** or the voltage value detected by the voltage sensor **116** and data about the failure of the first cooler **400** based on the corresponding current value or voltage value.

[0259] Thereafter, the at least one processor **351** may input the current value detected by the current sensor **115** or the voltage value detected by the voltage sensor **116** to the trained AI model to determine whether a failure has occurred in the first cooler **400** by the AI model.

[0260] In a case where the number of failure occurrences of the first cooler **400** is less than a reference number (No in operation **1005**), the at least one processor **351** may stop the operation of the first cooler **400** (**1007**). The reference number of failure occurrences may be set to perform separate control in a case where the failure of the first cooler **400** continues, and may be, for example, 3 times.

[0261] The at least one processor **351** may restart the first cooler **400** (**1011**), after a second period of time has elapsed after stopping the operation of the first cooler **400** (Yes in operation **1009**). Here, the second period of time may be, for example, 10 minutes.

[0262] Because the number of times the first cooler **400** has been detected as faulty is less than the reference number, the first cooler **400** may be stopped for a relatively short period of time (e.g., 10 minutes) and restarted to confirm again whether the first cooler **400** is operating normally.

[0263] As such, in a case where the first cooler **400** is detected as faulty, the first cooler **400** may be stopped and restarted to detect whether a failure has occurred in the first cooler **400** again based on a current flowing through the thermoelectric element **530**.

[0264] In a case where the current flowing through the thermoelectric element **530** continues to be outside the reference current range for the predetermined period of time even though the above-described process is repeated several times, the at least one processor **351** may perform separate control.

[0265] That is, in a case where the failure of the first cooler **400** has occurred more than the reference number of times (e.g., three times) (Yes in operation **1005**), the at least one processor **351** may stop the operation of the first cooler **400** (**1013**), and may restart the first cooler **400** (**1017**) in response to the elapse of a third period of time longer than the second period of time (Yes in operation **1015**).

[0266] Here, the third period of time may be an appropriate time for stopping the operation of the first cooler **400** whose failure has been repeated several times for a relatively long period of time, and may be, for example, 24 hours.

[0267] As such, whether a failure has occurred in the first cooler **400** may be detected by detecting the current flowing through the thermoelectric element **530**, and control may be performed based on the detection, thereby allowing normal operation of the first cooler **400**.

[0268] The at least one processor **351** may detect that the failure has been resolved in a case where the detected current after detecting the failure in the first cooler **400** is not outside the reference current range for the first period of time or longer.

[0269] FIG. **11** and FIG. **12** are flowcharts illustrating operations of a refrigerator depending on whether a failure has occurred in a thermoelectric cooler based on a temperature detection result according to an embodiment of the disclosure.

[0270] As another example of detecting an occurrence of a failure in the first cooler **400**, the at least one processor **351** may detect that a failure has occurred in the first cooler **400** in a case where a temperature detected by the heat absorption sensor **850** or the heat dissipation sensor **860** is outside a reference range.

[0271] Referring to FIG. **11**, in a case where the heat absorption sensor **850** detects a temperature of air absorbed by the thermoelectric element **530** and the detected temperature value is determined to be outside a reference temperature range for a fourth period of time (Yes in operation **1101**), the at least one processor **351** may detect that a failure has occurred in the first cooler **400**.

[0272] Here, the fourth period of time may be set to an appropriate period of time for detecting whether a failure has occurred in the thermoelectric element **530**, and may be, for example, a period of time for detecting a sensing value **50** times. The reference temperature range may also be an appropriate temperature value range for detecting whether a failure has occurred in the thermoelectric element **530**, and may be, for example, −25° C. to 95° C.

[0273] That is, in a case where the temperature of the air absorbed by the thermoelectric element **530** is detected to be outside the range of −25° C. to 95° C. for the fourth period of time, the at least one processor **351** may detect that a failure has occurred in the first cooler **400**.

[0274] The at least one processor **351** may reduce a maximum operating voltage of the first cooler **400** in a case where the temperature detected by the heat absorption sensor **850** is outside the reference temperature range for the fourth period of time.

[0275] For example, when the temperature detected by the heat absorption sensor **850** is significantly low, ice and the like may accumulate, lowering an operating efficiency. Accordingly, the at least one processor **351** may reduce the maximum operating voltage by adjusting an operating rate of the thermoelectric element **530**. For example, the maximum operating voltage of the first cooler **400** may be reduced to a first voltage (**1103**), and the first voltage may be 22 V.

[0276] Referring to FIG. **12**, in a case where the heat dissipation sensor **860** detects a temperature of air discharged by the thermoelectric element **530** and the detected temperature value is determined to be outside a reference temperature range for the fourth period of time (Yes in operation **1201**), the at least one processor **351** may detect that a failure has occurred in the first cooler **400**.

[0277] Here, the reference temperature range may be the same as the reference temperature range for detecting a failure based on the detection result by the heat absorption sensor **850**. For example, the reference temperature range may be −25° C. to 95° C. In addition, because a detection value by the heat absorption sensor **850** and a detection value by the heat dissipation sensor **860** are generally different, the reference temperature ranges may be set differently.

[0278] That is, in a case where the reference temperature ranges are set to be the same and the temperature of the air discharged by the thermoelectric element **530** is detected to be outside the range of −25° C. to 95° C. for the fourth period of time, the at least one processor **351** may detect that a failure has occurred in the first cooler **400**.

[0279] The at least one processor **351** may reduce the maximum operating voltage of the first cooler **400** in a case where the temperature detected by the heat dissipation sensor **860** is outside the reference temperature range for the fourth period of time.

[0280] For example, when the temperature detected by the heat dissipation sensor **860** is significantly high, the thermoelectric element **530** may be overheated, and thus to prevent the above, the at least one processor **351** may reduce the maximum operating voltage of the first cooler **400** by adjusting an operating rate of the thermoelectric element **530**. For example, the maximum operating voltage of the first cooler **400** may be reduced to a second voltage lower than the first voltage (**1203**), and the second voltage may be 18 V.

[0281] As such, whether a failure has occurred in the first cooler **400** may be detected by detecting the temperature of the air absorbed or discharged by the thermoelectric element **530**, and control may be performed based on the detection, thereby allowing efficient operation.

[0282] The at least one processor **351** may detect that the failure has been resolved in a case where the detected temperature after detecting the failure in the first cooler **400** is not outside the reference temperature range for the fourth period of time or longer.

[0283] In the above-described embodiment, it has been described that the heat absorption sensor **850** and the heat dissipation sensor **860** are included in the first cooler **400**, but the heat absorption sensor **850** and the heat dissipation sensor **860** may be configured to detect the temperature of air absorbed or discharged by the thermoelectric element **530** as a separate configuration of the first cooler **400**.

[0284] FIG. **13** is a flowchart illustrating operations of a refrigerator depending on whether a failure has occurred in a thermoelectric cooler based on a fan rotation speed according to an embodiment of the disclosure.

[0285] As still another example of detecting an occurrence of a failure in the first cooler **400**, the at least one processor **351** may detect that a failure has occurred in the first cooler **400** in a case where a rotation speed of the heat dissipation fan **600** or the cooling fan **800** is less than a reference rotation speed.

[0286] Referring to FIG. **13**, in a case where the rotation speed of the heat dissipation fan **600** or the cooling fan **800** is detected and the detected rotation speed is determined to be less than the reference rotation speed for a fifth period of time (Yes in operation **1301**), the at least one processor **351** may detect that a failure has occurred in the first cooler **400**.

[0287] Here, the fifth period of time may be set to an appropriate period of time for detecting whether a failure has occurred in the thermoelectric element **530**, and may be, for example, 10 minutes. The reference rotation speed may also be an appropriate rotation speed value for detecting whether a failure has occurred in the thermoelectric element **530**, and may be, for example, 400 revolutions per minute (RPM).

[0288] That is, in a case where the rotation speed of the heat dissipation fan **600** or the cooling fan **800** is detected to be less than 400 RPM for the fifth period of time, the at least one processor **351** may detect that a failure has occurred in the first cooler **400**.

[0289] The at least one processor **351** may stop the operation of the thermoelectric element **530** while maintaining the operation of the heat dissipation fan **600** or the cooling fan **800** (**1303**), in a case where the rotation speed of the heat dissipation fan **600** or the cooling fan **800** is less than the reference rotation speed for the fifth period of time.

[0290] As such, whether a failure has occurred in the first cooler **400** may be detected by detecting the rotation speed of the heat dissipation fan **600** or the cooling fan **800**, and control may be performed based on the detection, thereby allowing efficient operation.

[0291] The at least one processor **351** may detect that the failure has been resolved in a case where the rotation speed of the heat dissipation fan **600** or the cooling fan **800** is determined to be greater than or equal to the reference rotation speed for a sixth period of time or longer.

[0292] According to an embodiment of the disclosure, a refrigerator may include: a main body including a storage compartment; a first cooler including a thermoelectric element, a heat dissipation fan, and a cooling fan, and configured to cool the storage compartment; a second cooler including a compressor and an evaporator fan, and configured to cool the storage compartment, and at least one processor configured to increase an operating speed of the compressor and the evaporator fan of the second cooler based on a failure in the first cooler.

[0293] According to the disclosure, a proper temperature of the refrigerator may be maintained and efficient operation may be performed by detecting various types of failure of a thermoelectric cooler and performing control based on the type of failure.

[0294] In addition, a temperature of the refrigerator may be maintained by increasing an operating intensity of a refrigeration cycle device in response to a failure of a thermoelectric cooler.

[0295] The at least one processor may be configured to increase or decrease a target temperature of the storage compartment based on the failure in the first cooler.

[0296] The refrigerator may further include a current sensor configured to detect a current flowing through the thermoelectric element, and the failure in the first cooler may include the current detected by the current sensor being outside a reference current range for a first period of time.

[0297] The at least one processor may be configured to stop an operation of the first cooler in response to the current detected by the current sensor being outside the reference current range for the first period of time, and restart the first cooler in response to an elapse of a second period of time.

[0298] The at least one processor may be configured to stop the operation of the first cooler for a third period of time longer than the second period of time based on the failure in the first cooler occurring more than a reference number of times.

[0299] The refrigerator may further include a voltage sensor configured to detect a voltage applied to the thermoelectric element, and the failure in the first cooler may include the voltage detected by the voltage sensor being outside a reference voltage range for a first period of time.

[0300] The at least one processor may be configured to stop an operation of the first cooler in response to the voltage detected by the voltage sensor being outside the reference voltage range for the first period of time, and restart the first cooler in response to an elapse of a second period of time.

[0301] The first cooler may further include: a heat absorption sensor configured to detect a temperature of air absorbed by the thermoelectric element; and a heat dissipation sensor configured to detect a temperature of air discharged by the thermoelectric element, and the failure in the first cooler may include the temperature detected by the heat absorption sensor or the heat dissipation sensor being outside a reference temperature range for a fourth period of time.

[0302] The at least one processor may be configured to reduce a maximum operating voltage of the first cooler in response to the temperature detected by the heat absorption sensor or the heat dissipation sensor being outside the reference temperature range for the fourth period of time.

[0303] The at least one processor may be configured to reduce the maximum operating voltage of the first cooler to a first voltage, in response to the temperature detected by the heat absorption sensor being outside the reference temperature range for the fourth period of time, and reduce the maximum operating voltage of the first cooler to a second voltage lower than the first voltage, in response to the temperature detected by the heat dissipation sensor being outside the reference temperature range for the fourth period of time

[0304] The failure in the first cooler may include a rotation speed of the heat dissipation fan or the cooling fan being less than a reference rotation speed for a fifth period of time.

[0305] The at least one processor may be configured to maintain an operation of the heat dissipation fan or the cooling fan and stop an operation of the thermoelectric element, in response to the rotation speed of the heat dissipation fan or the cooling fan being less than the reference rotation speed for the fifth period of time

[0306] According to an embodiment of the disclosure, in a method for controlling a refrigerator including a main body including a storage compartment, a first cooler including a thermoelectric element, a heat dissipation fan, and a cooling fan and configured to cool the storage compartment, and a second cooler including a compressor and an evaporator fan and configured to cool the storage compartment, the method may include: detecting whether a failure occurs in the first cooler; and increasing an operating speed of the compressor and the evaporator fan of the second cooler based on the failure in the first cooler.

[0307] The method may further include increasing or decreasing a target temperature of the storage compartment based on the failure in the first cooler.

[0308] The refrigerator may further include a current sensor configured to detect a current flowing through the thermoelectric element, and the failure in the first cooler may include the current detected by the current sensor being outside a reference current range for a first period of time.

[0309] The method may further include stopping an operation of the first cooler in response to the current detected by the current sensor being outside the reference current range for the first period of time, and restarting the first cooler in response to an elapse of a second period of time.

[0310] The method may further include stopping the operation of the first cooler for a third period of time longer than the second period of time based on the failure in the first cooler occurring more than a reference number of times.

[0311] The first cooler may further include: a heat absorption sensor configured to detect a temperature of air absorbed by the thermoelectric element; and a heat dissipation sensor configured to detect a temperature of air discharged by the thermoelectric element, and the failure in the first cooler may include the temperature detected by the heat absorption sensor or the heat dissipation sensor being outside a reference temperature range for a fourth period of time.

[0312] The method may further include reducing a maximum operating voltage of the first cooler in response to the temperature detected by the heat absorption sensor or the heat dissipation sensor being outside the reference temperature range for the fourth period of time.

[0313] The reducing of the maximum operating voltage of the first cooler may include: reducing the maximum operating voltage of the first cooler to a first voltage, in response to the temperature detected by the heat absorption sensor being outside the reference temperature range for the fourth period of time, and reducing the maximum operating voltage of the first cooler to a second voltage lower than the first voltage, in response to the temperature detected by the heat dissipation sensor being outside the reference temperature range for the fourth period of time.

[0314] The failure in the first cooler may include a rotation speed of the heat dissipation fan or the cooling fan being less than a reference rotation speed for a fifth period of time.

[0315] The method may further include maintaining an operation of the heat dissipation fan or the cooling fan and stopping an operation of the thermoelectric element, in response to the rotation speed of the heat dissipation fan or the cooling fan being less than the reference rotation speed for the fifth period of time.

[0316] According to the disclosure, the refrigerator and the method for controlling the same may maintain a proper temperature of the refrigerator and operate efficiently by detecting various types of failure of a thermoelectric cooler and performing control based on the type of failure.

[0317] In addition, the refrigerator and the method for controlling the same may maintain a temperature of the refrigerator by increasing an operating intensity of a refrigeration cycle device in response to a failure of a thermoelectric cooler.

[0318] Meanwhile, the disclosed embodiments may be implemented in the form of a recording medium that stores instructions executable by a computer. The instructions may be stored in the form of program codes, and when executed by a processor, the instructions may create a program module to perform operations of the disclosed embodiments. The recording medium may be implemented as a computer-readable recording medium.

[0319] The computer-readable recording medium may include all kinds of recording media storing instructions that can be interpreted by a computer. For example, the computer-readable recording medium may be read only memory (ROM), random access memory (RAM), a magnetic tape, a magnetic disc, a flash memory, an optical data storage device, etc.

[0320] Also, the computer-readable recording medium may be provided in the form of a non-transitory storage medium, wherein the term 'non-transitory storage medium' simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium. For example, a 'non-transitory storage medium' may include a buffer in which data is temporarily stored.

[0321] According to an embodiment of the disclosure, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloadable or uploadable) online via an application store (e.g., Play Store™) or between two user devices (e.g., smart phones) directly. When distributed online, at least part of the computer program product (e.g., a downloadable app) may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as a memory of the manufacturer's server, a server of the application store, or a relay server.

[0322] Although embodiments of the disclosure have been described with reference to the accompanying drawings, a person having ordinary skilled in the art will appreciate that other specific modifications may be easily made without departing from the technical spirit or essential features of the disclosure. Accordingly, the foregoing embodiments should be regarded as illustrative rather than limiting in all aspects.

1. A refrigerator, comprising:

a main body including a storage compartment;

a first cooler configured to cool the storage compartment, and including:

a thermoelectric element,
a heat dissipation fan, and
a cooling fan;
a second cooler configured to cool the storage compartment, and including:
a compressor, and
an evaporator fan; and
at least one processor configured to, based on a failure in the first cooler, increase an operating speed of the compressor and the evaporator fan of the second cooler.

2. The refrigerator of claim **1**, wherein
the at least one processor is configured to:
based on the failure in the first cooler, increase or decrease a target temperature of the storage compartment.

3. The refrigerator of claim **1**, further comprising:
a current sensor configured to detect a current flowing through the thermoelectric element,
wherein the failure in the first cooler includes the current detected by the current sensor being outside a reference current range for a predetermined first period of time.

4. The refrigerator of claim **3**, wherein
the at least one processor is configured to:
based on the current detected by the current sensor being outside the reference current range for the predetermined first period of time, stop an operation of the first cooler, and
based on an elapse of a predetermined second period of time, restart the operation of the first cooler.

5. The refrigerator of claim **4**, wherein
the at least one processor is configured to:
based on the failure in the first cooler occurring more than a reference number of times, stop the operation of the first cooler for a predetermined third period of time that is longer than the predetermined second period of time.

6. The refrigerator of claim **1**, further comprising:
a voltage sensor configured to detect a voltage applied to the thermoelectric element,
wherein the failure in the first cooler includes the voltage detected by the voltage sensor being outside a reference voltage range for a predetermined first period of time.

7. The refrigerator of claim **6**, wherein
the at least one processor is configured to:
based on the voltage detected by the voltage sensor being outside the reference voltage range for the predetermined first period of time, stop an operation of the first cooler, and
based on an elapse of a predetermined second period of time, restart the operation of the first cooler.

8. The refrigerator of claim **1**, wherein
the first cooler includes:
a heat absorption sensor configured to detect a temperature of air absorbed by the thermoelectric element, and
a heat dissipation sensor configured to detect a temperature of air discharged by the thermoelectric element, and
the failure in the first cooler includes the temperature detected by the heat absorption sensor or the heat dissipation sensor being outside a reference temperature range for a predetermined period of time.

9. The refrigerator of claim **8**, wherein
the at least one processor is configured to:
based on the temperature detected by the heat absorption sensor or the heat dissipation sensor being outside the reference temperature range for the predetermined period of time, reduce a maximum operating voltage of the first cooler.

10. The refrigerator of claim **9**, wherein
the at least one processor is configured to:
based on the temperature detected by the heat absorption sensor being outside the reference temperature range for the predetermined period of time, reduce the maximum operating voltage of the first cooler to a first voltage, and
based on the temperature detected by the heat dissipation sensor being outside the reference temperature range for the predetermined period of time, reduce the maximum operating voltage of the first cooler to a second voltage.

11. The refrigerator of claim **1**, wherein
the failure in the first cooler includes a rotation speed of the heat dissipation fan or the cooling fan being less than a reference rotation speed for a predetermined period of time.

12. The refrigerator of claim **11**, wherein
the at least one processor is configured to, based on the rotation speed of the heat dissipation fan or the cooling fan being less than the reference rotation speed for the predetermined period of time:
maintain an operation of the heat dissipation fan or the cooling fan, and
stop an operation of the thermoelectric element.

13. A method of controlling a refrigerator including a main body including a storage compartment, a first cooler configured to cool the storage compartment and including a thermoelectric element, a heat dissipation fan, and a cooling fan, and a second cooler configured to cool the storage compartment and including a compressor and an evaporator fan, the method comprising:
detecting whether a failure occurs in the first cooler; and
based on the failure in the first cooler being detected, increasing an operating speed of the compressor and the evaporator fan of the second cooler.

14. The method of claim **13**, further comprising:
based on the failure in the first cooler being detected, increasing or decreasing a target temperature of the storage compartment.

15. The method of claim **13**, the refrigerator including a current sensor configured to detect a current flowing through the thermoelectric element, wherein
the failure in the first cooler being detected includes the current detected by the current sensor being outside a reference current range for a predetermined period of time.

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